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	Building A Seismic Design	Date		10022500446

Building A Seismic Design

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1 Effective Seismic Weight Calculation

We calculate the effective seismic weight with unit weights and square footages.

1.1 Unit Weights

Weights will be assigned with area unit weights.

$$U_{corridor} = 0.035 \text{ ksf} \quad U_{floor} = 0.035 \text{ ksf} \quad U_{roof} = 0.025 \text{ ksf} \quad U_{extwall} = 0.012 \text{ ksf} \quad (\text{ASCE 7-22 Table C3.1-1a})$$

1.2 Building Geometry

The lengths and widths of the weight areas are consistent across levels 2 and up.

$$L_{floor} = 230 \text{ ft} \quad w_{floor} = 30.75 \text{ ft} \quad L_{corridor} = 230 \text{ ft} \quad w_{corridor} = 5.5 \text{ ft} \quad w_{building} = 67 \text{ ft}$$

The wall heights vary across the levels.

$$h_{wall,1} = 11.5 \text{ ft} \quad h_{wall,2} = 10.5 \text{ ft} \quad h_{wall,3} = 10.5 \text{ ft} \quad h_{wall,4} = 10 \text{ ft}$$

The building perimeter will be used to calculate exterior wall weights.

$$\begin{aligned} L_{jog} &= 5 \cdot \text{ft} = 5 \cdot \text{ft} & &= 5 \text{ ft} \\ P_{wall} &= (2 \cdot L_{floor}) + (2 \cdot w_{building}) + (2 \cdot L_{jog}) \\ &= (2 \cdot 230 \text{ ft}) + (2 \cdot 67 \text{ ft}) + (2 \cdot 5 \text{ ft}) \\ &= 604 \text{ ft} \end{aligned}$$

Areas can now be calculated. Note that the areas of the walls are the actual wall areas per level; they have not been distributed to each floor based on tributary area, yet (divided in half to the floor above and below).

$$\begin{aligned} A_{floor} &= 2 \cdot (L_{floor} \cdot w_{floor}) = 2 \cdot (230 \text{ ft} \cdot 30.75 \text{ ft}) = 1.414e + 04 \text{ ft}^2 \\ A_{corridor} &= L_{corridor} \cdot w_{corridor} = 230 \text{ ft} \cdot 5.5 \text{ ft} = 1265 \text{ ft}^2 \\ A_{level} &= L_{floor} \cdot w_{building} = 230 \text{ ft} \cdot 67 \text{ ft} = 1.541e + 04 \text{ ft}^2 \\ A_{wall,1} &= h_{wall,1} \cdot P_{wall} = 11.5 \text{ ft} \cdot 604 \text{ ft} = 6946 \text{ ft}^2 \\ A_{wall,2} &= h_{wall,2} \cdot P_{wall} = 10.5 \text{ ft} \cdot 604 \text{ ft} = 6342 \text{ ft}^2 \\ A_{wall,3} &= h_{wall,3} \cdot P_{wall} = 10.5 \text{ ft} \cdot 604 \text{ ft} = 6342 \text{ ft}^2 \\ A_{wall,4} &= h_{wall,4} \cdot P_{wall} = 10 \text{ ft} \cdot 604 \text{ ft} = 6040 \text{ ft}^2 \end{aligned}$$

1.3 Component Weights

The typical floor and corridor weights are

$$\begin{aligned} W_{floor,typ} &= U_{floor} \cdot A_{floor} = 0.035 \text{ ksf} \cdot 1.414e + 04 \text{ ft}^2 = 495.1 \text{ kip} \\ W_{corridor,typ} &= U_{corridor} \cdot A_{corridor} = 0.035 \text{ ksf} \cdot 1265 \text{ ft}^2 = 44.28 \text{ kip} \end{aligned}$$

This applies to levels 2 and up.

$$\begin{aligned} W_{floor,2} &= 495.1 \text{ kip} & W_{floor,3} &= 495.1 \text{ kip} & W_{floor,4} &= 495.1 \text{ kip} \\ W_{corridor,2} &= 44.28 \text{ kip} & W_{corridor,3} &= 44.28 \text{ kip} & W_{corridor,4} &= 44.28 \text{ kip} \end{aligned}$$



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The roof weight is

$$W_{roof} = U_{roof} \cdot A_{level} = 0.025 \text{ ksf} \cdot 1.541e + 04 \text{ ft}^2 = 385.2 \text{ kip}$$

The exterior walls are divided in half to the floor above and below.

$$W_{wall,2} = U_{extwall} \cdot \left(\frac{A_{wall,2}}{2} + \frac{A_{wall,1}}{2} \right) = 0.012 \text{ ksf} \cdot \left(\frac{6342 \text{ ft}^2}{2} + \frac{6946 \text{ ft}^2}{2} \right) = 79.73 \text{ kip}$$

$$W_{wall,3} = U_{extwall} \cdot \left(\frac{A_{wall,3}}{2} + \frac{A_{wall,2}}{2} \right) = 0.012 \text{ ksf} \cdot \left(\frac{6342 \text{ ft}^2}{2} + \frac{6342 \text{ ft}^2}{2} \right) = 76.1 \text{ kip}$$

$$W_{wall,4} = U_{extwall} \cdot \left(\frac{A_{wall,4}}{2} + \frac{A_{wall,3}}{2} \right) = 0.012 \text{ ksf} \cdot \left(\frac{6040 \text{ ft}^2}{2} + \frac{6342 \text{ ft}^2}{2} \right) = 74.29 \text{ kip}$$

$$W_{wall,roof} = U_{extwall} \cdot \left(\frac{A_{wall,4}}{2} \right) = 0.012 \text{ ksf} \cdot \left(\frac{6040 \text{ ft}^2}{2} \right) = 36.24 \text{ kip}$$

1.4 Level Weights

Each level's weight is

$$W_{level,2} = W_{floor,2} + W_{corridor,2} + W_{wall,2} = 495.1 \text{ kip} + 44.28 \text{ kip} + 79.73 \text{ kip} = 619.1 \text{ kip}$$

$$W_{level,3} = W_{floor,3} + W_{corridor,3} + W_{wall,3} = 495.1 \text{ kip} + 44.28 \text{ kip} + 76.1 \text{ kip} = 615.5 \text{ kip}$$

$$W_{level,4} = W_{floor,4} + W_{corridor,4} + W_{wall,4} = 495.1 \text{ kip} + 44.28 \text{ kip} + 74.29 \text{ kip} = 613.6 \text{ kip}$$

$$W_{level,roof} = W_{roof} + W_{wall,roof} = 385.2 \text{ kip} + 36.24 \text{ kip} = 421.5 \text{ kip}$$

1.5 Total Weight

The total weight of the structure is

$$\begin{aligned} W_{total} &= W_{level,2} + W_{level,3} + W_{level,4} + W_{level,roof} \\ &= 619.1 \text{ kip} + 615.5 \text{ kip} + 613.6 \text{ kip} + 421.5 \text{ kip} \\ &= 2270 \text{ kip} \end{aligned}$$

2 Seismic Equivalent Lateral Force (ELF) Analysis

We will be using the equivalent lateral force method prescribed by ASCE 7-22 Section 12.8. We will envelope the forces with a fully flexible diaphragm assumption and a fully rigid diaphragm assumption.

2.1 Seismic Parameters

Seismic design parameters are obtained from the ASCE Hazard Tool and ASCE 7-22 Sec. 11 and 12.

For latitude = 45.501 and longitude = -122.834:

$$I_e = 1 \quad S_S = 0.97 \quad S_1 = 0.38 \quad S_{MS} = 1.2 \quad S_{M1} = 0.82 \quad S_{DS} = 0.8 \quad S_{D1} = 0.55 \quad SDC = D \quad T_L = 16$$

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2.2 Base Shear

2.3 Structural seismic parameters

The lateral force resisting systems in each direction (E-W/x and N-S/y) have the following seismic design coefficients.

$$h_{top,roof} = 58.25 \text{ ft}$$

$$h_{bottom,roof} = 42.5 \text{ ft}$$

$$h_n = 50.38 \text{ ft (ASCE 7-22 Sec. 11.2)}$$

$$h_{n,check} = 50.4 \text{ ft} \leq 65 \text{ ft} \quad \text{OK} \checkmark$$

The lateral system in the E-W/x direction is: A16 = Light-frame (wood) walls sheathed with wood structural panels rated for shear resistance

$$R_x = 6.5 \quad \Omega_{0,x} = 3 \quad C_{d,x} = 4 \quad \rho_x = 1$$

The lateral system in the N-S/y direction is: A16 = Light-frame (wood) walls sheathed with wood structural panels rated for shear resistance

$$R_y = 6.5 \quad \Omega_{0,y} = 3 \quad C_{d,y} = 4 \quad \rho_y = 1$$

We will approximate the building period with

$$T_a = C_t h_n^x \quad (\text{ASCE 7-22 Eq. 12.8-8})$$

$$c_{t,x} = 0.02 \quad x_x = 0.75$$

$$c_{t,y} = 0.02 \quad x_y = 0.75$$

$$T_x = 0.3782 \quad T_y = 0.3782$$

And now, we can calculate the base shear, V

$$C_{s,x} = 0.1231 \quad C_{s,y} = 0.1231$$

$$V_x = 279.3 \quad V_y = 279.3$$

2.4 Distribute the seismic forces

With the base shear calculated, we must now distribute the base shear on each level.

Calculating for each floor leads to the following [Table 1](#) and [Table 2](#).

Table 1: Seismic Forces Calculation Table - EW

Level	h_{floor} [ft]	w_x [kip]	h_x [ft]	$w_x h_x^k$	C_{vx}	F_x [kip]	$\sum F_x$ [kip]	V_x [kip]	OTM [kip-ft]
5	0.0	421	42.5	17913	0.306	85.5	85.5	0.0	0.0
4	10.0	614	32.5	19943	0.341	95.2	181	85.5	855
3	10.5	615	22.0	13540	0.231	64.6	245	181	2753
2	10.5	619	11.5	7119	0.122	34.0	279	245	5329
1	11.5	0.0	0.0	0.0	0.0	0.0	279	279	8541

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Table 2: Seismic Forces Calculation Table - NS

Level	h_{floor} [ft]	w_x [kip]	h_x [ft]	$w_x h_x^k$	C_{vx}	F_x [kip]	ΣF_x [kip]	V_x [kip]	OTM [kip-ft]
5	0.0	421	42.5	17913	0.306	85.5	85.5	0.0	0.0
4	10.0	614	32.5	19943	0.341	95.2	181	85.5	855
3	10.5	615	22.0	13540	0.231	64.6	245	181	2753
2	10.5	619	11.5	7119	0.122	34.0	279	245	5329
1	11.5	0.0	0.0	0.0	0.0	0.0	279	279	8541

3 Diaphragm and Shear Wall Design

3.1 Redundancy Factor

The redundancy factor, ρ , is either 1.3 or 1.0 according to ASCE 7-22 12.3.4.

No single story resists more than 35% of the shear force (see C_{vx} in Table 1 and Table 2), so all values of ρ are 1.0, initially.

3.2 Shear Wall Design

We will envelope the forces with a fully flexible diaphragm assumption and a fully rigid diaphragm assumption.

The shear wall naming goes *[E-W line]-[N-S line intersection of wall left edge]*. For example, wall “A-C-A-3.8” is the wall on line “A-C” whose left edge begins at “A-4.2”. If there is only one wall on the line, then the name is just the grid line.

3.2.1 Flexible Diaphragm Assumption

We will uniformly smear the shear forces across the diaphragms based on tributary areas of each level. This removes some nuance in the distribution of floor weight distribution, but other areas of the design are conservative and make up for this assumption.

$$f_{p,x} = F_x / A_{level}$$

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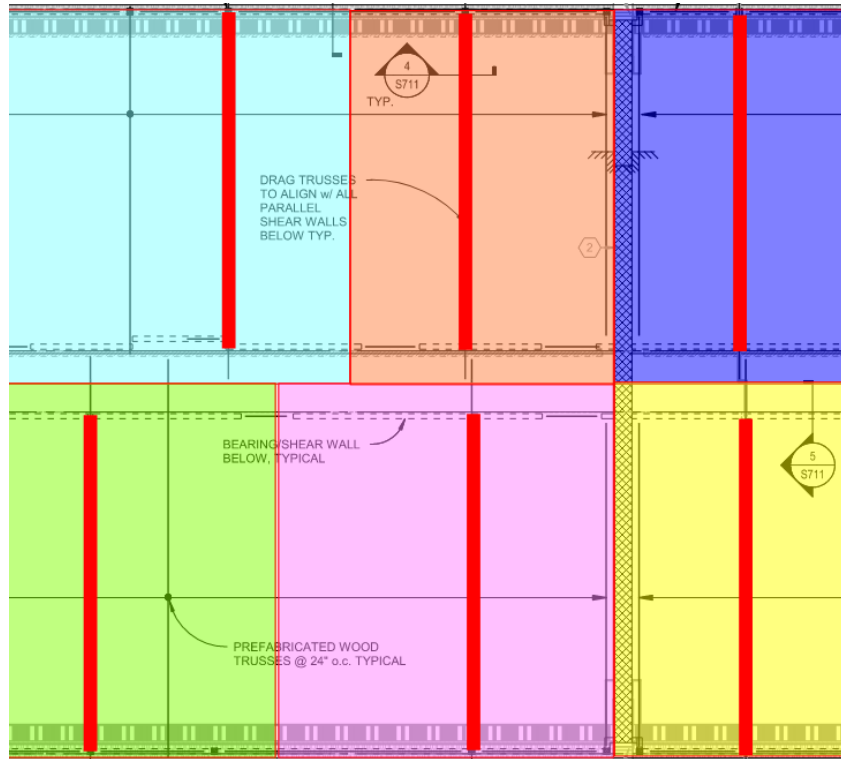


Figure 1: Shear walls (thick red lines) in plan view with their (colored) tributary areas

Shear wall shear forces are tabulated below.

Shear wall sheathing will be assumed to have the following properties:

1. WSP Sheathing grade
2. 15/32" thickness
3. 10d common nails
4. Plywood panels
5. Framing uses Douglas-Fir-Larch (no specific gravity adjustment)

Shear capacity and stiffness values come from **AWC SDPWS 2021 Table 4.3A**.

Notes:

- Cumulative shear force, $F_{x,cum}$, is cumulative from the Roof downwards.



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Table 3: Shear wall forces x - Level 4

Wall	w_{trib} [psf]	A [ft ²]	F_x [lbf]	$F_{x,cum}$ [lbf]	l_{wall} [ft]	v [plf]	$0.7v$ [lbf]	v_{cap} [plf]	sides	s_{nail} [in]
A-B-A-1	5.55	1106	6663	6663	20.0	356	233	311	1.00	6.00
A-B-A-2.5	5.55	720	4331	4331	13.0	333	233	311	1.00	6.00
A-B-A-3.5	5.55	561	2998	2998	9.00	333	233	311	1.00	6.00
A-B-A-4	5.55	653	3998	3998	12.0	333	233	311	1.00	6.00
A-B-A-4.8	5.55	536	3555	3555	10.7	333	233	311	1.00	6.00
A-B-A-5.5	5.55	1139	9662	9662	29.0	362	233	311	1.00	6.00
A-B-A-6.5	5.55	963	4331	4331	13.0	333	233	311	1.00	6.00
A-B-A-7.9	5.55	938	3888	3888	11.7	333	233	311	1.00	6.00
A-B-A-8.8	5.55	1089	3332	3332	10.0	333	233	311	1.00	6.00
A-C-A-1.8	5.55	1563	4740	4740	16.0	317	207	311	1.00	6.00
A-C-A-3.2	5.55	469	2767	2767	9.34	296	207	311	1.00	6.00
A-C-A-3.8	5.55	720	4962	4962	16.8	319	207	311	1.00	6.00
A-C-A-4.6	5.55	972	4740	4740	16.0	317	207	311	1.00	6.00
A-C-A-5.5	5.55	681	4443	4443	15.0	314	207	311	1.00	6.00
A-C-A-6.5	5.55	1022	7702	7702	26.0	328	207	311	1.00	6.00
A-C-A-6.9	5.55	536	3407	3407	11.5	296	207	311	1.00	6.00
A-C-A-8.5	5.55	787	2888	2888	9.75	296	207	311	1.00	6.00
A-C-A-9.2	5.55	955	7109	7109	24.0	327	207	311	1.00	6.00

Table 4: Shear wall forces x - Level 3

Wall	w_{trib} [psf]	A [ft ²]	F_x [lbf]	$F_{x,cum}$ [lbf]	l_{wall} [ft]	v [plf]	$0.7v$ [lbf]	v_{cap} [plf]	sides	s_{nail} [in]
A-B-A-1	6.18	1106	7418	14081	20.0	732	493	600	1.00	3.00
A-B-A-2.5	6.18	720	4822	9153	13.0	704	493	600	1.00	3.00
A-B-A-3.5	6.18	561	3338	6337	9.00	704	493	600	1.00	3.00
A-B-A-4	6.18	653	4451	8449	12.0	704	493	600	1.00	3.00
A-B-A-4.8	6.18	536	3958	7512	10.7	704	493	600	1.00	3.00
A-B-A-5.5	6.18	1139	10756	20418	29.0	737	493	600	1.00	3.00
A-B-A-6.5	6.18	963	4822	9153	13.0	704	493	600	1.00	3.00
A-B-A-7.9	6.18	938	4329	8216	11.7	704	493	600	1.00	3.00
A-B-A-8.8	6.18	1089	3709	7041	10.0	704	493	600	1.00	3.00
A-C-A-1.8	6.18	1563	5277	10016	16.0	645	438	461	1.00	4.00
A-C-A-3.2	6.18	469	3080	5847	9.34	626	438	461	1.00	4.00
A-C-A-3.8	6.18	720	5524	10486	16.8	646	438	461	1.00	4.00
A-C-A-4.6	6.18	972	5277	10016	16.0	645	438	461	1.00	4.00
A-C-A-5.5	6.18	681	4947	9390	15.0	641	438	461	1.00	4.00
A-C-A-6.5	6.18	1022	8575	16277	26.0	718	438	461	1.00	4.00
A-C-A-6.9	6.18	536	3793	7199	11.5	626	438	461	1.00	4.00
A-C-A-8.5	6.18	787	3216	6104	9.75	626	438	461	1.00	4.00
A-C-A-9.2	6.18	955	7915	15025	24.0	716	438	461	1.00	4.00



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Table 5: Shear wall forces x - Level 2

Wall	w_{trib} [psf]	A [ft ²]	F_x [lbf]	$F_{x,cum}$ [lbf]	l_{wall} [ft]	v [plf]	$0.7v$ [lbf]	v_{cap} [plf]	sides	s_{nail} [in]
A-B-A-1	4.19	1106	5036	19118	20.0	956	669	770	1.00	2.00
A-B-A-2.5	4.19	720	3274	12427	13.0	956	669	770	1.00	2.00
A-B-A-3.5	4.19	561	2266	8603	9.00	956	669	770	1.00	2.00
A-B-A-4	4.19	653	3022	11471	12.0	956	669	770	1.00	2.00
A-B-A-4.8	4.19	536	2687	10199	10.7	956	669	770	1.00	2.00
A-B-A-5.5	4.19	1139	7303	27721	29.0	1300	669	770	1.00	2.00
A-B-A-6.5	4.19	963	3274	12427	13.0	956	669	770	1.00	2.00
A-B-A-7.9	4.19	938	2939	11155	11.7	956	669	770	1.00	2.00
A-B-A-8.8	4.19	1089	2518	9559	10.0	956	669	770	1.00	2.00
A-C-A-1.8	4.19	1563	3582	13599	16.0	850	595	600	1.00	3.00
A-C-A-3.2	4.19	469	2091	7938	9.34	850	595	600	1.00	3.00
A-C-A-3.8	4.19	720	3750	14236	16.8	850	595	600	1.00	3.00
A-C-A-4.6	4.19	972	3582	13599	16.0	850	595	600	1.00	3.00
A-C-A-5.5	4.19	681	3359	12749	15.0	850	595	600	1.00	3.00
A-C-A-6.5	4.19	1022	5822	22098	26.0	973	595	600	1.00	3.00
A-C-A-6.9	4.19	536	2575	9774	11.5	850	595	600	1.00	3.00
A-C-A-8.5	4.19	787	2183	8287	9.75	850	595	600	1.00	3.00
A-C-A-9.2	4.19	955	5374	20398	24.0	970	595	600	1.00	3.00

Table 6: Shear wall forces x - Level 1

Wall	w_{trib} [psf]	A [ft ²]	F_x [lbf]	$F_{x,cum}$ [lbf]	l_{wall} [ft]	v [plf]	$0.7v$ [lbf]	v_{cap} [plf]	sides	s_{nail} [in]
A-B-A-1	2.21	1106	2648	21766	20.0	1346	762	770	1.00	2.00
A-B-A-2.5	2.21	720	1721	14148	13.0	1088	762	770	1.00	2.00
A-B-A-3.5	2.21	561	1192	9795	9.00	1088	762	770	1.00	2.00
A-B-A-4	2.21	653	1589	13060	12.0	1088	762	770	1.00	2.00
A-B-A-4.8	2.21	536	1413	11612	10.7	1088	762	770	1.00	2.00
A-B-A-5.5	2.21	1139	3840	31561	29.0	1395	762	770	1.00	2.00
A-B-A-6.5	2.21	963	1721	14148	13.0	1088	762	770	1.00	2.00
A-B-A-7.9	2.21	938	1545	12700	11.7	1088	762	770	1.00	2.00
A-B-A-8.8	2.21	1089	1324	10883	10.0	1088	762	770	1.00	2.00
A-C-A-1.8	2.21	1563	1884	15483	16.0	968	677	770	1.00	2.00
A-C-A-3.2	2.21	469	1100	9038	9.34	968	677	770	1.00	2.00
A-C-A-3.8	2.21	720	1972	16208	16.8	968	677	770	1.00	2.00
A-C-A-4.6	2.21	972	1884	15483	16.0	968	677	770	1.00	2.00
A-C-A-5.5	2.21	681	1766	14515	15.0	968	677	770	1.00	2.00
A-C-A-6.5	2.21	1022	3061	25159	26.0	1246	677	770	1.00	2.00
A-C-A-6.9	2.21	536	1354	11128	11.5	968	677	770	1.00	2.00
A-C-A-8.5	2.21	787	1148	9435	9.75	968	677	770	1.00	2.00
A-C-A-9.2	2.21	955	2826	23224	24.0	1238	677	770	1.00	2.00



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Table 7: Shear wall forces y - Level 4

Wall	w_{trib} [psf]	A [ft ²]	F_x [lbf]	$F_{x,cum}$ [lbf]	l_{wall} [ft]	v [plf]	$0.7v$ [lbf]	v_{cap} [plf]	sides	s_{nail} [in]
A-2	5.55	1369	7599	7599	29.8	255	179	311	1.00	6.00
A-2.4	5.55	1572	8722	8722	29.8	293	205	311	1.00	6.00
A-3	5.55	593	3292	3292	29.8	142	77.5	311	1.00	6.00
A-3.5	5.55	972	5391	5391	29.8	202	127	311	1.00	6.00
A-3.7	5.55	441	2448	2448	20.0	142	85.7	311	1.00	6.00
A-4	5.55	505	2804	2804	20.0	156	98.1	311	1.00	6.00
A-5	5.55	770	4276	4276	29.8	166	101	311	1.00	6.00
A-5.1	5.55	988	5484	5484	29.8	201	129	311	1.00	6.00
A-6	5.55	988	5484	5484	29.8	198	129	311	1.00	6.00
A-6.1	5.55	826	4586	4586	29.8	173	108	311	1.00	6.00
A-6.8	5.55	906	5027	5027	29.8	188	118	311	1.00	6.00
A-7	5.55	1012	5616	5616	29.8	206	132	311	1.00	6.00
A-7.8	5.55	641	3555	3555	20.0	183	124	311	1.00	6.00
A-8	5.55	794	4407	4407	29.8	173	104	311	1.00	6.00
A-8.2	5.55	448	2486	2486	20.0	144	87.0	311	1.00	6.00
A-8.9	5.55	1353	7506	7506	29.8	252	177	311	1.00	6.00
A-9	5.55	1231	6832	6832	29.8	240	161	311	1.00	6.00

Table 8: Shear wall forces y - Level 3

Wall	w_{trib} [psf]	A [ft ²]	F_x [lbf]	$F_{x,cum}$ [lbf]	l_{wall} [ft]	v [plf]	$0.7v$ [lbf]	v_{cap} [plf]	sides	s_{nail} [in]
A-2	6.18	1369	8460	16059	29.8	557	378	461	1.00	4.00
A-2.4	6.18	1572	9710	18432	29.8	620	434	461	1.00	4.00
A-3	6.18	593	3665	6957	29.8	282	164	311	1.00	6.00
A-3.5	6.18	972	6002	11393	29.8	417	268	311	1.00	6.00
A-3.7	6.18	441	2725	5173	20.0	299	181	311	1.00	6.00
A-4	6.18	505	3122	5926	20.0	330	207	311	1.00	6.00
A-5	6.18	770	4760	9036	29.8	339	213	311	1.00	6.00
A-5.1	6.18	988	6106	11590	29.8	415	273	311	1.00	6.00
A-6	6.18	988	6106	11590	29.8	406	273	311	1.00	6.00
A-6.1	6.18	826	5105	9691	29.8	355	228	311	1.00	6.00
A-6.8	6.18	906	5597	10624	29.8	387	250	311	1.00	6.00
A-7	6.18	1012	6252	11868	29.8	424	279	311	1.00	6.00
A-7.8	6.18	641	3958	7514	20.0	388	263	311	1.00	6.00
A-8	6.18	794	4907	9314	29.8	352	219	311	1.00	6.00
A-8.2	6.18	448	2768	5255	20.0	303	184	311	1.00	6.00
A-8.9	6.18	1353	8356	15862	29.8	549	373	461	1.00	4.00
A-9	6.18	1231	7606	14438	29.8	527	340	461	1.00	4.00

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Table 9: Shear wall forces y - Level 2

Wall	w_{trib} [psf]	A [ft ²]	F_x [lbf]	$F_{x,cum}$ [lbf]	l_{wall} [ft]	v [plf]	$0.7v$ [lbf]	v_{cap} [plf]	sides	s_{nail} [in]
A-2	4.19	1369	5744	21802	29.8	733	513	600	1.00	3.00
A-2.4	4.19	1572	6593	25025	29.8	841	589	600	1.00	3.00
A-3	4.19	593	2488	9445	29.8	420	222	311	1.00	6.00
A-3.5	4.19	972	4075	15468	29.8	556	364	461	1.00	4.00
A-3.7	4.19	441	1850	7023	20.0	385	246	311	1.00	6.00
A-4	4.19	505	2119	8045	20.0	427	282	311	1.00	6.00
A-5	4.19	770	3232	12268	29.8	509	289	311	1.00	6.00
A-5.1	4.19	988	4145	15735	29.8	554	370	461	1.00	4.00
A-6	4.19	988	4145	15735	29.8	549	370	461	1.00	4.00
A-6.1	4.19	826	3466	13157	29.8	535	310	311	1.00	6.00
A-6.8	4.19	906	3800	14424	29.8	515	339	461	1.00	4.00
A-7	4.19	1012	4245	16113	29.8	567	379	461	1.00	4.00
A-7.8	4.19	641	2687	10201	20.0	535	357	461	1.00	4.00
A-8	4.19	794	3331	12646	29.8	529	298	311	1.00	6.00
A-8.2	4.19	448	1879	7134	20.0	389	250	311	1.00	6.00
A-8.9	4.19	1353	5673	21535	29.8	724	507	600	1.00	3.00
A-9	4.19	1231	5164	19602	29.8	693	461	600	1.00	3.00

Table 10: Shear wall forces y - Level 1

Wall	w_{trib} [psf]	A [ft ²]	F_x [lbf]	$F_{x,cum}$ [lbf]	l_{wall} [ft]	v [plf]	$0.7v$ [lbf]	v_{cap} [plf]	sides	s_{nail} [in]
A-2	2.21	1369	3020	24822	29.8	834	584	600	1.00	3.00
A-2.4	2.21	1572	3466	28491	29.8	958	670	770	1.00	2.00
A-3	2.21	593	1308	10754	29.8	479	253	311	1.00	6.00
A-3.5	2.21	972	2143	17611	29.8	637	414	461	1.00	4.00
A-3.7	2.21	441	973	7996	20.0	505	280	311	1.00	6.00
A-4	2.21	505	1114	9160	20.0	506	321	461	1.00	4.00
A-5	2.21	770	1699	13967	29.8	515	329	461	1.00	4.00
A-5.1	2.21	988	2180	17914	29.8	637	422	461	1.00	4.00
A-6	2.21	988	2180	17914	29.8	632	422	461	1.00	4.00
A-6.1	2.21	826	1822	14979	29.8	542	352	461	1.00	4.00
A-6.8	2.21	906	1998	16422	29.8	592	386	461	1.00	4.00
A-7	2.21	1012	2232	18345	29.8	654	432	461	1.00	4.00
A-7.8	2.21	641	1413	11614	20.0	615	406	461	1.00	4.00
A-8	2.21	794	1752	14397	29.8	536	339	461	1.00	4.00
A-8.2	2.21	448	988	8122	20.0	514	284	311	1.00	6.00
A-8.9	2.21	1353	2983	24518	29.8	824	577	600	1.00	3.00
A-9	2.21	1231	2715	22317	29.8	788	525	600	1.00	3.00

Check aspect ratios We will also check the aspect ratios. Ensuring an aspect ratio under 2 means no strength reduction is necessary.

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$$\text{ratio}_{\max} = 1.28 < 2 \quad \text{OK} \checkmark$$

∴ All aspect ratios are OK

3.2.2 Rigid Diaphragm Assumption

With a rigid diaphragm assumption, we now must calculate the following in each direction:

1. The relative stiffnesses of the shear walls to obtain a center of rigidity.
2. The center of mass of the diaphragm with the center of rigidity to obtain an eccentricity.

The relative stiffnesses of the shear walls can be calculated by using the deflection equation from NDS SDPWS Sec. 4.3.4

$$k = \frac{F}{\delta}, \text{ assume } F = \text{induced load demand in the shear wall}$$

$$\delta_{sw} = \frac{8vh^3}{EAb} + \frac{vh}{1000G_a} + \frac{h\Delta_A}{b} \quad (\text{SDPWS Eq. 4.3-1})$$

where

- b = the shear wall length [ft]
- Δ_A = vertical deformation of the wall overturning anchorage system + vertical compression deformation [in]
- E = modulus of elasticity of end posts [psi]
- A = area of end post cross-section [in²]
- G_a = apparent shear wall shear stiffness from nail slip and panel shear deformation [kip/in]
- h = shear wall height [ft]
- v = unit shear force induced by the design load [plf]
- δ_{sw} = maximum shear wall deflection determined by elastic analysis [in]

Note that for the following calculations, we will conservatively use the unit shear force induced by the nominal load, not the design load, for the value of v . This is because the stiffnesses obtained from this use will allow us to use the stiffnesses for deflection limits and story drift calculations

We will assume that the end posts are (2) 2x4 Douglas Fir No. 2; this means:

$$E_{\text{endpost}} = 1600000 \text{ (psi)}$$

$$A_{\text{endpost}} = 10.5 \text{ (in}^2\text{)}$$

We will assume a continuous tiedown rod system. A conservative first-pass value for Δ_A can be:

$$\Delta_A = 0.2 \text{ (in)}$$

Shear Wall Rigidities With the deflections of each shear wall, along with the force demand of each shear wall, we can obtain the stiffnesses (and therefore, relative stiffnesses) of all the walls.

Each relative stiffness is relative to the level. The sum of all relative stiffnesses on each level is 1.0.

$$k = \frac{F_x}{\delta_{sw}}$$

$$R = \frac{k_{\text{wall}}}{\sum k_{\text{wall}}}$$

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Table 11: Wall rigidities x - Level 4

Wall	F_x [lbf]	δ_{sw} [in]	k [lbf/in]	R [-]
A-B-A-1	6663	0.0302	220610	0.0802
A-B-A-2.5	4331	0.0337	128639	0.0468
A-B-A-3.5	2998	0.0398	75302	0.0274
A-B-A-4	3998	0.0348	114811	0.0417
A-B-A-4.8	3555	0.0367	96887	0.0352
A-B-A-5.5	9662	0.0278	347665	0.126
A-B-A-6.5	4331	0.0337	128639	0.0468
A-B-A-7.9	3888	0.0352	110311	0.0401
A-B-A-8.8	3332	0.0378	88091	0.0320
A-C-A-1.8	4740	0.0301	157533	0.0573
A-C-A-3.2	2767	0.0367	75315	0.0274
A-C-A-3.8	4962	0.0297	167102	0.0607
A-C-A-4.6	4740	0.0301	157533	0.0573
A-C-A-5.5	4443	0.0307	144919	0.0527
A-C-A-6.5	7702	0.0264	291205	0.106
A-C-A-6.9	3407	0.0331	102771	0.0374
A-C-A-8.5	2888	0.0359	80379	0.0292
A-C-A-9.2	7109	0.0270	263571	0.0958

Table 12: Wall rigidities x - Level 3

Wall	F_x [lbf]	δ_{sw} [in]	k [lbf/in]	R [-]
A-B-A-1	14081	0.0441	319146	0.0816
A-B-A-2.5	9153	0.0484	189212	0.0484
A-B-A-3.5	6337	0.0555	114251	0.0292
A-B-A-4	8449	0.0497	169987	0.0435
A-B-A-4.8	7512	0.0519	144869	0.0370
A-B-A-5.5	20418	0.0411	496200	0.127
A-B-A-6.5	9153	0.0484	189212	0.0484
A-B-A-7.9	8216	0.0502	163703	0.0418
A-B-A-8.8	7041	0.0532	132447	0.0339
A-C-A-1.8	10016	0.0460	217922	0.0557
A-C-A-3.2	5847	0.0540	108204	0.0277
A-C-A-3.8	10486	0.0455	230513	0.0589
A-C-A-4.6	10016	0.0460	217922	0.0557
A-C-A-5.5	9390	0.0466	201530	0.0515
A-C-A-6.5	16277	0.0410	396526	0.101
A-C-A-6.9	7199	0.0499	144159	0.0369
A-C-A-8.5	6104	0.0531	114904	0.0294
A-C-A-9.2	15025	0.0416	361022	0.0923



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Table 13: Wall rigidities x - Level 2

Wall	F_x [lbf]	δ_{sw} [in]	k [lbf/in]	R [-]
A-B-A-1	19118	0.0473	404088	0.0837
A-B-A-2.5	12427	0.0532	233561	0.0484
A-B-A-3.5	8603	0.0607	141756	0.0294
A-B-A-4	11471	0.0546	210055	0.0435
A-B-A-4.8	10199	0.0569	179307	0.0371
A-B-A-5.5	27721	0.0415	667340	0.138
A-B-A-6.5	12427	0.0532	233561	0.0484
A-B-A-7.9	11155	0.0551	202367	0.0419
A-B-A-8.8	9559	0.0583	164083	0.0340
A-C-A-1.8	13599	0.0525	258931	0.0536
A-C-A-3.2	7938	0.0621	127917	0.0265
A-C-A-3.8	14236	0.0519	274196	0.0568
A-C-A-4.6	13599	0.0525	258931	0.0536
A-C-A-5.5	12749	0.0534	238695	0.0494
A-C-A-6.5	22098	0.0455	485826	0.101
A-C-A-6.9	9774	0.0578	169238	0.0350
A-C-A-8.5	8287	0.0611	135639	0.0281
A-C-A-9.2	20398	0.0460	443030	0.0918

Table 14: Wall rigidities x - Level 1

Wall	F_x [lbf]	δ_{sw} [in]	k [lbf/in]	R [-]
A-B-A-1	21766	0.0476	457256	0.0905
A-B-A-2.5	14148	0.0651	217186	0.0430
A-B-A-3.5	9795	0.0739	132468	0.0262
A-B-A-4	13060	0.0668	195528	0.0387
A-B-A-4.8	11612	0.0695	167166	0.0331
A-B-A-5.5	31561	0.0447	706110	0.140
A-B-A-6.5	14148	0.0651	217186	0.0430
A-B-A-7.9	12700	0.0674	188440	0.0373
A-B-A-8.8	10883	0.0711	153108	0.0303
A-C-A-1.8	15483	0.0559	276729	0.0548
A-C-A-3.2	9038	0.0671	134708	0.0267
A-C-A-3.8	16208	0.0552	293371	0.0581
A-C-A-4.6	15483	0.0559	276729	0.0548
A-C-A-5.5	14515	0.0570	254691	0.0504
A-C-A-6.5	25159	0.0454	554271	0.110
A-C-A-6.9	11128	0.0621	179300	0.0355
A-C-A-8.5	9435	0.0660	143021	0.0283
A-C-A-9.2	23224	0.0460	504912	0.0999



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Table 15: Wall rigidities γ - Level 4

Wall	F_x [lbf]	δ_{sw} [in]	k [lbf/in]	R [-]
A-2	7599	0.0211	359337	0.0775
A-2.4	8722	0.0234	372030	0.0802
A-3	3292	0.0142	231172	0.0499
A-3.5	5391	0.0179	300804	0.0649
A-3.7	2448	0.0171	143274	0.0309
A-4	2804	0.0179	156311	0.0337
A-5	4276	0.0157	271968	0.0586
A-5.1	5484	0.0178	307794	0.0664
A-6	5484	0.0177	310298	0.0669
A-6.1	4586	0.0161	284193	0.0613
A-6.8	5027	0.0170	294928	0.0636
A-7	5616	0.0181	309921	0.0668
A-7.8	3555	0.0196	181465	0.0391
A-8	4407	0.0162	272828	0.0588
A-8.2	2486	0.0172	144424	0.0311
A-8.9	7506	0.0210	358162	0.0772
A-9	6832	0.0202	338245	0.0729

Table 16: Wall rigidities γ - Level 3

Wall	F_x [lbf]	δ_{sw} [in]	k [lbf/in]	R [-]
A-2	16059	0.0354	453508	0.0802
A-2.4	18432	0.0387	475931	0.0841
A-3	6957	0.0239	290895	0.0514
A-3.5	11393	0.0326	349478	0.0618
A-3.7	5173	0.0281	183917	0.0325
A-4	5926	0.0301	196547	0.0347
A-5	9036	0.0276	327713	0.0579
A-5.1	11590	0.0325	356812	0.0631
A-6	11590	0.0319	363301	0.0642
A-6.1	9691	0.0286	338366	0.0598
A-6.8	10624	0.0307	346316	0.0612
A-7	11868	0.0331	359049	0.0635
A-7.8	7514	0.0339	221679	0.0392
A-8	9314	0.0284	327847	0.0580
A-8.2	5255	0.0284	185184	0.0327
A-8.9	15862	0.0350	453226	0.0801
A-9	14438	0.0338	427018	0.0755

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Table 17: Wall rigidities γ - Level 2

Wall	F_x [lbf]	δ_{sw} [in]	k [lbf/in]	R [-]
A-2	21802	0.0408	534794	0.0803
A-2.4	25025	0.0459	544973	0.0818
A-3	9445	0.0328	288211	0.0432
A-3.5	15468	0.0354	437459	0.0656
A-3.7	7023	0.0337	208592	0.0313
A-4	8045	0.0364	221095	0.0332
A-5	12268	0.0329	373034	0.0560
A-5.1	15735	0.0353	446115	0.0669
A-6	15735	0.0350	449862	0.0675
A-6.1	13157	0.0343	384104	0.0576
A-6.8	14424	0.0332	434964	0.0653
A-7	16113	0.0360	448127	0.0672
A-7.8	10201	0.0375	271729	0.0408
A-8	12646	0.0339	372490	0.0559
A-8.2	7134	0.0339	210219	0.0315
A-8.9	21535	0.0403	533892	0.0801
A-9	19602	0.0389	504384	0.0757

Table 18: Wall rigidities γ - Level 1

Wall	F_x [lbf]	δ_{sw} [in]	k [lbf/in]	R [-]
A-2	24822	0.0502	494277	0.0768
A-2.4	28491	0.0483	590013	0.0916
A-3	10754	0.0344	312300	0.0485
A-3.5	17611	0.0436	403477	0.0627
A-3.7	7996	0.0396	201925	0.0314
A-4	9160	0.0396	231110	0.0359
A-5	13967	0.0365	382417	0.0594
A-5.1	17914	0.0436	410467	0.0637
A-6	17914	0.0434	413127	0.0642
A-6.1	14979	0.0381	393061	0.0610
A-6.8	16422	0.0410	400286	0.0622
A-7	18345	0.0446	411033	0.0638
A-7.8	11614	0.0461	251979	0.0391
A-8	14397	0.0378	381253	0.0592
A-8.2	8122	0.0401	202473	0.0314
A-8.9	24518	0.0497	493493	0.0766
A-9	22317	0.0478	467221	0.0726

Center of Rigidity and Center of Mass We can find the center of rigidity by taking the weighted sum of all wall stiffnesses and their locations relative to a datum.



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$$\bar{x}_r = \frac{\sum k_{ns}x}{\sum k_{ns}}, \bar{y}_r = \frac{\sum k_{ew}y}{\sum k_{ew}}$$

Note: each wall is assumed to have zero rigidity in the direction perpendicular to them.

An accidental torsion eccentricity of 5% must be added/subtracted.

$$CM_{x,a,+} = CM_x + X_{plan} \cdot 0.05 = 117.5 + 235 \cdot 0.05 = 129.2$$

$$CM_{x,a,-} = CM_x - X_{plan} \cdot 0.05 = 117.5 - 235 \cdot 0.05 = 105.8$$

$$CM_{y,a,+} = CM_y + Y_{plan} \cdot 0.05 = 33.5 + 67 \cdot 0.05 = 36.85$$

$$CM_{y,a,-} = CM_y - Y_{plan} \cdot 0.05 = 33.5 - 67 \cdot 0.05 = 30.15$$

The eccentricity used is determined by the location of the wall in relation to the center of rigidity. Since the wall rotates about the center of rigidity, we want to position the center of mass to ensure each wall is getting the maximum possible force to be conservative. If the center of mass is positioned on the opposite side of the center of rigidity relative to the wall in question, that mass will *decrease* the shear experienced by that wall.

This means we always want the center of mass to be on the same side of the center of rigidity as the wall. The eccentricities shown in the following tables are based on the worst-case scenario. This is not immediately obvious due to all torques being given a positive value.

How the center of rigidity, center of mass, and eccentricity interact are illustrated in [Figure 2](#). The moments generated by the rotation of the center of mass around the center of rigidity differ based on the chosen eccentricity.

$$d_x = |x - CR_x| \text{ and } d_y = |y - CR_y|$$

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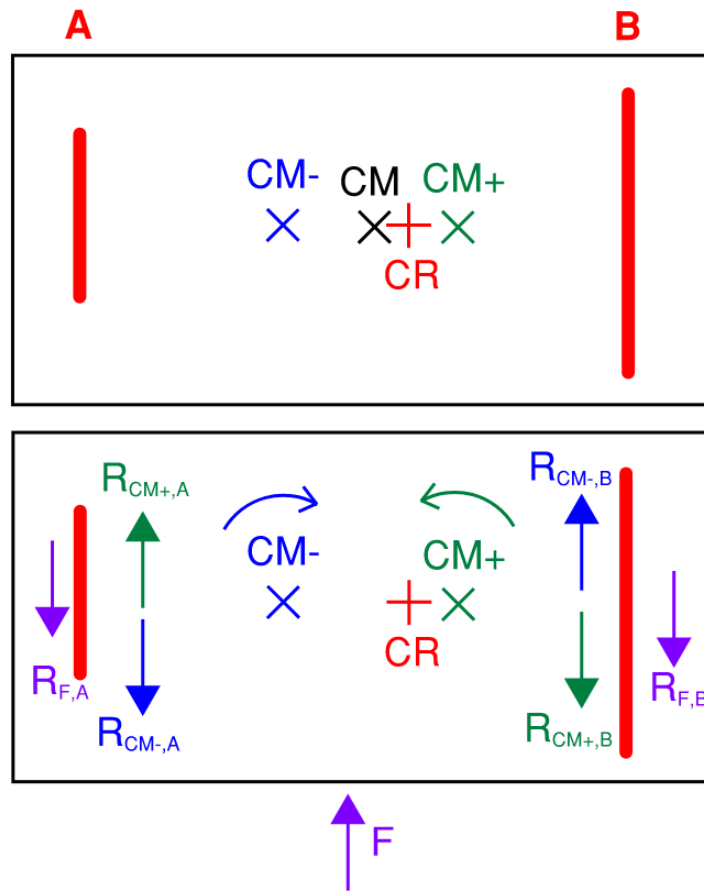


Figure 2: Moments applied to a rigid diaphragm

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Table 19: Center of Rigidity - Level 4

Wall	x [ft]	y [ft]	CR _x [ft]	d _x [ft]	e _x [ft]	CR _y [ft]	d _y [ft]	e _y [ft]
A-2	31.92	0.0	118.7	86.74	13.70	0.0	0.0	4.664
A-2.4	40.08	0.0	118.7	78.58	13.70	0.0	0.0	4.664
A-3	49.83	0.0	118.7	68.83	13.70	0.0	0.0	4.664
A-3.5	63.75	0.0	118.7	54.91	13.70	0.0	0.0	4.664
A-3.7	67.33	0.0	118.7	51.33	13.70	0.0	0.0	4.664
A-4	76.17	0.0	118.7	42.49	13.70	0.0	0.0	4.664
A-5	97.50	0.0	118.7	21.16	13.70	0.0	0.0	4.664
A-5.1	98.08	0.0	118.7	20.58	13.70	0.0	0.0	4.664
A-6	122.2	0.0	118.7	3.507	10.59	0.0	0.0	4.185
A-6.1	122.8	0.0	118.7	4.090	10.59	0.0	0.0	4.185
A-6.8	147.4	0.0	118.7	28.76	10.59	0.0	0.0	4.185
A-7	156.5	0.0	118.7	37.84	10.59	0.0	0.0	4.185
A-7.8	176.8	0.0	118.7	58.17	10.59	0.0	0.0	4.185
A-8	182.6	0.0	118.7	63.92	10.59	0.0	0.0	4.185
A-8.2	185.7	0.0	118.7	67.01	10.59	0.0	0.0	4.185
A-8.9	203.6	0.0	118.7	84.92	10.59	0.0	0.0	4.185
A-9	203.9	0.0	118.7	85.26	10.59	0.0	0.0	4.185
A-B-A-1	0.0	30.75	0.0	0.0	11.75	33.66	2.911	5.047
A-B-A-2.5	0.0	30.75	0.0	0.0	11.75	33.66	2.911	5.047
A-B-A-3.5	0.0	30.75	0.0	0.0	11.75	33.66	2.911	5.047
A-B-A-4	0.0	30.75	0.0	0.0	11.75	33.66	2.911	5.047
A-B-A-4.8	0.0	30.75	0.0	0.0	11.75	33.66	2.911	5.047
A-B-A-5.5	0.0	30.75	0.0	0.0	11.75	33.66	2.911	5.047
A-B-A-6.5	0.0	30.75	0.0	0.0	11.75	33.66	2.911	5.047
A-B-A-7.9	0.0	30.75	0.0	0.0	11.75	33.66	2.911	5.047
A-B-A-8.8	0.0	30.75	0.0	0.0	11.75	33.66	2.911	5.047
A-C-A-1.8	0.0	36.25	0.0	0.0	11.75	33.66	2.589	4.726
A-C-A-3.2	0.0	36.25	0.0	0.0	11.75	33.66	2.589	4.726
A-C-A-3.8	0.0	36.25	0.0	0.0	11.75	33.66	2.589	4.726
A-C-A-4.6	0.0	36.25	0.0	0.0	11.75	33.66	2.589	4.726
A-C-A-5.5	0.0	36.25	0.0	0.0	11.75	33.66	2.589	4.726
A-C-A-6.5	0.0	36.25	0.0	0.0	11.75	33.66	2.589	4.726
A-C-A-6.9	0.0	36.25	0.0	0.0	11.75	33.66	2.589	4.726
A-C-A-8.5	0.0	36.25	0.0	0.0	11.75	33.66	2.589	4.726
A-C-A-9.2	0.0	36.25	0.0	0.0	11.75	33.66	2.589	4.726



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Table 20: Center of Rigidity - Level 3

Wall	x [ft]	y [ft]	CR _x [ft]	d _x [ft]	e _x [ft]	CR _y [ft]	d _y [ft]	e _y [ft]
A-2	31.92	0.0	118.6	86.73	13.52	0.0	0.0	4.569
A-2.4	40.08	0.0	118.6	78.56	13.52	0.0	0.0	4.569
A-3	49.83	0.0	118.6	68.81	13.52	0.0	0.0	4.569
A-3.5	63.75	0.0	118.6	54.90	13.52	0.0	0.0	4.569
A-3.7	67.33	0.0	118.6	51.31	13.52	0.0	0.0	4.569
A-4	76.17	0.0	118.6	42.48	13.52	0.0	0.0	4.569
A-5	97.50	0.0	118.6	21.15	13.52	0.0	0.0	4.569
A-5.1	98.08	0.0	118.6	20.56	13.52	0.0	0.0	4.569
A-6	122.2	0.0	118.6	3.519	10.60	0.0	0.0	4.115
A-6.1	122.8	0.0	118.6	4.102	10.60	0.0	0.0	4.115
A-6.8	147.4	0.0	118.6	28.77	10.60	0.0	0.0	4.115
A-7	156.5	0.0	118.6	37.85	10.60	0.0	0.0	4.115
A-7.8	176.8	0.0	118.6	58.19	10.60	0.0	0.0	4.115
A-8	182.6	0.0	118.6	63.94	10.60	0.0	0.0	4.115
A-8.2	185.7	0.0	118.6	67.02	10.60	0.0	0.0	4.115
A-8.9	203.6	0.0	118.6	84.94	10.60	0.0	0.0	4.115
A-9	203.9	0.0	118.6	85.27	10.60	0.0	0.0	4.115
A-B-A-1	0.0	30.75	0.0	0.0	11.75	33.56	2.806	5.345
A-B-A-2.5	0.0	30.75	0.0	0.0	11.75	33.56	2.806	5.345
A-B-A-3.5	0.0	30.75	0.0	0.0	11.75	33.56	2.806	5.345
A-B-A-4	0.0	30.75	0.0	0.0	11.75	33.56	2.806	5.345
A-B-A-4.8	0.0	30.75	0.0	0.0	11.75	33.56	2.806	5.345
A-B-A-5.5	0.0	30.75	0.0	0.0	11.75	33.56	2.806	5.345
A-B-A-6.5	0.0	30.75	0.0	0.0	11.75	33.56	2.806	5.345
A-B-A-7.9	0.0	30.75	0.0	0.0	11.75	33.56	2.806	5.345
A-B-A-8.8	0.0	30.75	0.0	0.0	11.75	33.56	2.806	5.345
A-C-A-1.8	0.0	36.25	0.0	0.0	11.75	33.56	2.694	5.234
A-C-A-3.2	0.0	36.25	0.0	0.0	11.75	33.56	2.694	5.234
A-C-A-3.8	0.0	36.25	0.0	0.0	11.75	33.56	2.694	5.234
A-C-A-4.6	0.0	36.25	0.0	0.0	11.75	33.56	2.694	5.234
A-C-A-5.5	0.0	36.25	0.0	0.0	11.75	33.56	2.694	5.234
A-C-A-6.5	0.0	36.25	0.0	0.0	11.75	33.56	2.694	5.234
A-C-A-6.9	0.0	36.25	0.0	0.0	11.75	33.56	2.694	5.234
A-C-A-8.5	0.0	36.25	0.0	0.0	11.75	33.56	2.694	5.234
A-C-A-9.2	0.0	36.25	0.0	0.0	11.75	33.56	2.694	5.234

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Table 21: Center of Rigidity - Level 2

Wall	x [ft]	y [ft]	CR _x [ft]	d _x [ft]	e _x [ft]	CR _y [ft]	d _y [ft]	e _y [ft]
A-2	31.92	0.0	118.9	86.99	14.07	0.0	0.0	4.728
A-2.4	40.08	0.0	118.9	78.83	14.07	0.0	0.0	4.728
A-3	49.83	0.0	118.9	69.08	14.07	0.0	0.0	4.728
A-3.5	63.75	0.0	118.9	55.16	14.07	0.0	0.0	4.728
A-3.7	67.33	0.0	118.9	51.58	14.07	0.0	0.0	4.728
A-4	76.17	0.0	118.9	42.74	14.07	0.0	0.0	4.728
A-5	97.50	0.0	118.9	21.41	14.07	0.0	0.0	4.728
A-5.1	98.08	0.0	118.9	20.83	14.07	0.0	0.0	4.728
A-6	122.2	0.0	118.9	3.255	10.34	0.0	0.0	4.142
A-6.1	122.8	0.0	118.9	3.839	10.34	0.0	0.0	4.142
A-6.8	147.4	0.0	118.9	28.51	10.34	0.0	0.0	4.142
A-7	156.5	0.0	118.9	37.59	10.34	0.0	0.0	4.142
A-7.8	176.8	0.0	118.9	57.92	10.34	0.0	0.0	4.142
A-8	182.6	0.0	118.9	63.67	10.34	0.0	0.0	4.142
A-8.2	185.7	0.0	118.9	66.76	10.34	0.0	0.0	4.142
A-8.9	203.6	0.0	118.9	84.67	10.34	0.0	0.0	4.142
A-9	203.9	0.0	118.9	85.01	10.34	0.0	0.0	4.142
A-B-A-1	0.0	30.75	0.0	0.0	11.75	33.47	2.724	5.655
A-B-A-2.5	0.0	30.75	0.0	0.0	11.75	33.47	2.724	5.655
A-B-A-3.5	0.0	30.75	0.0	0.0	11.75	33.47	2.724	5.655
A-B-A-4	0.0	30.75	0.0	0.0	11.75	33.47	2.724	5.655
A-B-A-4.8	0.0	30.75	0.0	0.0	11.75	33.47	2.724	5.655
A-B-A-5.5	0.0	30.75	0.0	0.0	11.75	33.47	2.724	5.655
A-B-A-6.5	0.0	30.75	0.0	0.0	11.75	33.47	2.724	5.655
A-B-A-7.9	0.0	30.75	0.0	0.0	11.75	33.47	2.724	5.655
A-B-A-8.8	0.0	30.75	0.0	0.0	11.75	33.47	2.724	5.655
A-C-A-1.8	0.0	36.25	0.0	0.0	11.75	33.47	2.776	5.707
A-C-A-3.2	0.0	36.25	0.0	0.0	11.75	33.47	2.776	5.707
A-C-A-3.8	0.0	36.25	0.0	0.0	11.75	33.47	2.776	5.707
A-C-A-4.6	0.0	36.25	0.0	0.0	11.75	33.47	2.776	5.707
A-C-A-5.5	0.0	36.25	0.0	0.0	11.75	33.47	2.776	5.707
A-C-A-6.5	0.0	36.25	0.0	0.0	11.75	33.47	2.776	5.707
A-C-A-6.9	0.0	36.25	0.0	0.0	11.75	33.47	2.776	5.707
A-C-A-8.5	0.0	36.25	0.0	0.0	11.75	33.47	2.776	5.707
A-C-A-9.2	0.0	36.25	0.0	0.0	11.75	33.47	2.776	5.707

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Table 22: Center of Rigidity - Level 1

Wall	x [ft]	y [ft]	CR _x [ft]	d _x [ft]	e _x [ft]	CR _y [ft]	d _y [ft]	e _y [ft]
A-2	31.92	0.0	118.1	86.19	12.91	0.0	0.0	4.547
A-2.4	40.08	0.0	118.1	78.02	12.91	0.0	0.0	4.547
A-3	49.83	0.0	118.1	68.27	12.91	0.0	0.0	4.547
A-3.5	63.75	0.0	118.1	54.35	12.91	0.0	0.0	4.547
A-3.7	67.33	0.0	118.1	50.77	12.91	0.0	0.0	4.547
A-4	76.17	0.0	118.1	41.94	12.91	0.0	0.0	4.547
A-5	97.50	0.0	118.1	20.60	12.91	0.0	0.0	4.547
A-5.1	98.08	0.0	118.1	20.02	12.91	0.0	0.0	4.547
A-6	122.2	0.0	118.1	4.064	11.26	0.0	0.0	4.298
A-6.1	122.8	0.0	118.1	4.648	11.26	0.0	0.0	4.298
A-6.8	147.4	0.0	118.1	29.31	11.26	0.0	0.0	4.298
A-7	156.5	0.0	118.1	38.40	11.26	0.0	0.0	4.298
A-7.8	176.8	0.0	118.1	58.73	11.26	0.0	0.0	4.298
A-8	182.6	0.0	118.1	64.48	11.26	0.0	0.0	4.298
A-8.2	185.7	0.0	118.1	67.56	11.26	0.0	0.0	4.298
A-8.9	203.6	0.0	118.1	85.48	11.26	0.0	0.0	4.298
A-9	203.9	0.0	118.1	85.81	11.26	0.0	0.0	4.298
A-B-A-1	0.0	30.75	0.0	0.0	11.75	33.67	2.920	6.212
A-B-A-2.5	0.0	30.75	0.0	0.0	11.75	33.67	2.920	6.212
A-B-A-3.5	0.0	30.75	0.0	0.0	11.75	33.67	2.920	6.212
A-B-A-4	0.0	30.75	0.0	0.0	11.75	33.67	2.920	6.212
A-B-A-4.8	0.0	30.75	0.0	0.0	11.75	33.67	2.920	6.212
A-B-A-5.5	0.0	30.75	0.0	0.0	11.75	33.67	2.920	6.212
A-B-A-6.5	0.0	30.75	0.0	0.0	11.75	33.67	2.920	6.212
A-B-A-7.9	0.0	30.75	0.0	0.0	11.75	33.67	2.920	6.212
A-B-A-8.8	0.0	30.75	0.0	0.0	11.75	33.67	2.920	6.212
A-C-A-1.8	0.0	36.25	0.0	0.0	11.75	33.67	2.580	5.873
A-C-A-3.2	0.0	36.25	0.0	0.0	11.75	33.67	2.580	5.873
A-C-A-3.8	0.0	36.25	0.0	0.0	11.75	33.67	2.580	5.873
A-C-A-4.6	0.0	36.25	0.0	0.0	11.75	33.67	2.580	5.873
A-C-A-5.5	0.0	36.25	0.0	0.0	11.75	33.67	2.580	5.873
A-C-A-6.5	0.0	36.25	0.0	0.0	11.75	33.67	2.580	5.873
A-C-A-6.9	0.0	36.25	0.0	0.0	11.75	33.67	2.580	5.873
A-C-A-8.5	0.0	36.25	0.0	0.0	11.75	33.67	2.580	5.873
A-C-A-9.2	0.0	36.25	0.0	0.0	11.75	33.67	2.580	5.873

And now, we can get the forces in the shear walls using the rigid diaphragm assumption. These forces can be compared to the flexible shear wall forces obtained earlier, F_{flex} .

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Table 23: Shear wall forces with the rigid diaphragm assumption - Level 4

Wall	T_x [lbf-ft]	T_y [lbf-ft]	d_x [ft]	d_y [ft]	R [-]	F_t [lbf]	F_v [lbf]	F_{tot} [lbf]	F_{flex} [lbf]
A-2	398862	1171000.0	86.74	0.0	0.07749	777.8	6661	7599	7599
A-2.4	398862	1171000.0	78.58	0.0	0.08023	729.5	6897	8722	8722
A-3	398862	1171000.0	68.83	0.0	0.04985	358.0	3865	4223	3292
A-3.5	398862	1171000.0	54.91	0.0	0.06487	414.5	5608	6023	5391
A-3.7	398862	1171000.0	51.33	0.0	0.03090	183.9	2662	2845	2448
A-4	398862	1171000.0	42.49	0.0	0.03371	169.0	2954	3123	2804
A-5	398862	1171000.0	21.16	0.0	0.05865	137.0	4810	4947	4276
A-5.1	398862	1171000.0	20.58	0.0	0.06638	160.9	5810	5971	5484
A-6	357867	905615	3.507	0.0	0.06692	24.89	5876	5901	5484
A-6.1	357867	905615	4.090	0.0	0.06129	25.31	5124	5149	4586
A-6.8	357867	905615	28.76	0.0	0.06360	187.8	5406	5594	5027
A-7	357867	905615	37.84	0.0	0.06683	267.4	5852	6119	5616
A-7.8	357867	905615	58.17	0.0	0.03913	240.3	3421	3661	3555
A-8	357867	905615	63.92	0.0	0.05884	369.7	4789	5158	4407
A-8.2	357867	905615	67.01	0.0	0.03114	216.2	2672	2888	2486
A-8.9	357867	905615	84.92	0.0	0.07724	681.0	6640	7506	7506
A-9	357867	905615	85.26	0.0	0.07294	666.1	6469	7135	6832
A-B-A-1	431567	1005000.0	0.0	2.911	0.08018	40.99	7070	7111	6663
A-B-A-2.5	431567	1005000.0	0.0	2.911	0.04676	23.54	4059	4331	4331
A-B-A-3.5	431567	1005000.0	0.0	2.911	0.02737	13.78	2376	2998	2998
A-B-A-4	431567	1005000.0	0.0	2.911	0.04173	21.01	3623	3998	3998
A-B-A-4.8	431567	1005000.0	0.0	2.911	0.03522	17.73	3057	3555	3555
A-B-A-5.5	431567	1005000.0	0.0	2.911	0.1264	60.52	10437	10498	9662
A-B-A-6.5	431567	1005000.0	0.0	2.911	0.04676	23.54	4059	4331	4331
A-B-A-7.9	431567	1005000.0	0.0	2.911	0.04009	20.18	3481	3888	3888
A-B-A-8.8	431567	1005000.0	0.0	2.911	0.03202	16.12	2780	3332	3332
A-C-A-1.8	404106	1005000.0	0.0	2.589	0.05726	26.05	5050	5076	4740
A-C-A-3.2	404106	1005000.0	0.0	2.589	0.02737	12.26	2377	2767	2767
A-C-A-3.8	404106	1005000.0	0.0	2.589	0.06074	27.42	5316	5343	4962
A-C-A-4.6	404106	1005000.0	0.0	2.589	0.05726	26.05	5050	5076	4740
A-C-A-5.5	404106	1005000.0	0.0	2.589	0.05267	24.21	4693	4717	4443
A-C-A-6.5	404106	1005000.0	0.0	2.589	0.1058	43.79	8490	8534	7702
A-C-A-6.9	404106	1005000.0	0.0	2.589	0.03735	16.73	3243	3407	3407
A-C-A-8.5	404106	1005000.0	0.0	2.589	0.02922	13.08	2536	2888	2888
A-C-A-9.2	404106	1005000.0	0.0	2.589	0.09580	40.32	7817	7857	7109

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Table 24: Shear wall forces with the rigid diaphragm assumption - Level 3

Wall	T_x [lbf-ft]	T_y [lbf-ft]	d_x [ft]	d_y [ft]	R [-]	F_t [lbf]	F_v [lbf]	F_{tot} [lbf]	F_{flex} [lbf]
A-2	825773	2443000.0	86.73	0.0	0.08017	1664	14906	16570	16059
A-2.4	825773	2443000.0	78.56	0.0	0.08413	1541	15240	18432	18432
A-3	825773	2443000.0	68.81	0.0	0.05142	681.5	7696	8377	6957
A-3.5	825773	2443000.0	54.90	0.0	0.06178	819.0	11592	12411	11393
A-3.7	825773	2443000.0	51.31	0.0	0.03251	370.5	5610	5980	5173
A-4	825773	2443000.0	42.48	0.0	0.03475	342.4	6263	6605	5926
A-5	825773	2443000.0	21.15	0.0	0.05793	267.0	9809	10076	9036
A-5.1	825773	2443000.0	20.56	0.0	0.06308	318.6	12037	12356	11590
A-6	743729	1916000.0	3.519	0.0	0.06422	49.10	12037	12086	11590
A-6.1	743729	1916000.0	4.102	0.0	0.05982	50.03	10521	10571	9691
A-6.8	743729	1916000.0	28.77	0.0	0.06122	371.7	11146	11518	10624
A-7	743729	1916000.0	37.85	0.0	0.06347	530.5	12091	12622	11868
A-7.8	743729	1916000.0	58.19	0.0	0.03919	490.4	7271	7761	7514
A-8	743729	1916000.0	63.94	0.0	0.05796	722.0	9743	10465	9314
A-8.2	743729	1916000.0	67.02	0.0	0.03274	436.6	5621	6057	5255
A-8.9	743729	1916000.0	84.94	0.0	0.08012	1464	14874	16339	15862
A-9	743729	1916000.0	85.27	0.0	0.07549	1410	14263	15673	14438
A-B-A-1	965987	2123000.0	0.0	2.806	0.08159	92.81	14539	14632	14081
A-B-A-2.5	965987	2123000.0	0.0	2.806	0.04837	56.26	8813	9153	9153
A-B-A-3.5	965987	2123000.0	0.0	2.806	0.02921	33.97	5322	6337	6337
A-B-A-4	965987	2123000.0	0.0	2.806	0.04346	50.55	7918	8449	8449
A-B-A-4.8	965987	2123000.0	0.0	2.806	0.03703	43.08	6748	7512	7512
A-B-A-5.5	965987	2123000.0	0.0	2.806	0.1268	135.6	21241	21377	20418
A-B-A-6.5	965987	2123000.0	0.0	2.806	0.04837	56.26	8813	9153	9153
A-B-A-7.9	965987	2123000.0	0.0	2.806	0.04185	48.68	7625	8216	8216
A-B-A-8.8	965987	2123000.0	0.0	2.806	0.03386	39.38	6169	7041	7041
A-C-A-1.8	945803	2123000.0	0.0	2.694	0.05571	62.83	10250	10313	10016
A-C-A-3.2	945803	2123000.0	0.0	2.694	0.02766	30.89	5040	5847	5847
A-C-A-3.8	945803	2123000.0	0.0	2.694	0.05893	65.96	10761	10827	10486
A-C-A-4.6	945803	2123000.0	0.0	2.694	0.05571	62.83	10250	10313	10016
A-C-A-5.5	945803	2123000.0	0.0	2.694	0.05152	58.53	9549	9608	9390
A-C-A-6.5	945803	2123000.0	0.0	2.694	0.1014	113.7	18545	18659	16277
A-C-A-6.9	945803	2123000.0	0.0	2.694	0.03685	41.16	6715	7199	7199
A-C-A-8.5	945803	2123000.0	0.0	2.694	0.02937	32.81	5352	6104	6104
A-C-A-9.2	945803	2123000.0	0.0	2.694	0.09229	104.6	17070	17174	15025



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Table 25: Shear wall forces with the rigid diaphragm assumption - Level 2

Wall	T_x [lbf-ft]	T_y [lbf-ft]	d_x [ft]	d_y [ft]	R [-]	F_t [lbf]	F_v [lbf]	F_{tot} [lbf]	F_{flex} [lbf]
A-2	1160000.0	3453000.0	86.99	0.0	0.08025	2298	19507	21805	21802
A-2.4	1160000.0	3453000.0	78.83	0.0	0.08178	2122	19876	25025	25025
A-3	1160000.0	3453000.0	69.08	0.0	0.04325	1069	11423	12491	9445
A-3.5	1160000.0	3453000.0	55.16	0.0	0.06564	1150	15392	16541	15468
A-3.7	1160000.0	3453000.0	51.58	0.0	0.03130	502.1	7189	7691	7023
A-4	1160000.0	3453000.0	42.74	0.0	0.03318	466.8	8064	8531	8045
A-5	1160000.0	3453000.0	21.41	0.0	0.05598	427.0	14727	15154	12268
A-5.1	1160000.0	3453000.0	20.83	0.0	0.06694	452.4	16039	16492	15735
A-6	1016000.0	2537000.0	3.255	0.0	0.06751	62.81	16264	16327	15735
A-6.1	1016000.0	2537000.0	3.839	0.0	0.05764	72.17	15849	15921	13157
A-6.8	1016000.0	2537000.0	28.51	0.0	0.06527	500.7	14807	15308	14424
A-7	1016000.0	2537000.0	37.59	0.0	0.06725	720.4	16156	16876	16113
A-7.8	1016000.0	2537000.0	57.92	0.0	0.04078	688.6	10021	10709	10201
A-8	1016000.0	2537000.0	63.67	0.0	0.05590	1106	14644	15750	12646
A-8.2	1016000.0	2537000.0	66.76	0.0	0.03155	570.5	7203	7774	7134
A-8.9	1016000.0	2537000.0	84.67	0.0	0.08012	1956	19472	21535	21535
A-9	1016000.0	2537000.0	85.01	0.0	0.07569	1888	18726	20614	19602
A-B-A-1	1387000.0	2883000.0	0.0	2.724	0.08369	125.1	18661	19118	19118
A-B-A-2.5	1387000.0	2883000.0	0.0	2.724	0.04837	78.08	11644	12427	12427
A-B-A-3.5	1387000.0	2883000.0	0.0	2.724	0.02936	47.39	7067	8603	8603
A-B-A-4	1387000.0	2883000.0	0.0	2.724	0.04350	70.22	10473	11471	11471
A-B-A-4.8	1387000.0	2883000.0	0.0	2.724	0.03714	59.94	8940	10199	10199
A-B-A-5.5	1387000.0	2883000.0	0.0	2.724	0.1382	251.0	37438	37689	27721
A-B-A-6.5	1387000.0	2883000.0	0.0	2.724	0.04837	78.08	11644	12427	12427
A-B-A-7.9	1387000.0	2883000.0	0.0	2.724	0.04191	67.65	10089	11155	11155
A-B-A-8.8	1387000.0	2883000.0	0.0	2.724	0.03398	54.85	8181	9559	9559
A-C-A-1.8	1400000.0	2883000.0	0.0	2.776	0.05363	88.23	12909	13599	13599
A-C-A-3.2	1400000.0	2883000.0	0.0	2.776	0.02649	43.59	6377	7938	7938
A-C-A-3.8	1400000.0	2883000.0	0.0	2.776	0.05679	93.41	13667	14236	14236
A-C-A-4.6	1400000.0	2883000.0	0.0	2.776	0.05363	88.23	12909	13599	13599
A-C-A-5.5	1400000.0	2883000.0	0.0	2.776	0.04943	81.34	11900	12749	12749
A-C-A-6.5	1400000.0	2883000.0	0.0	2.776	0.1006	171.8	25139	25311	22098
A-C-A-6.9	1400000.0	2883000.0	0.0	2.776	0.03505	57.67	8438	9774	9774
A-C-A-8.5	1400000.0	2883000.0	0.0	2.776	0.02809	46.22	6762	8287	8287
A-C-A-9.2	1400000.0	2883000.0	0.0	2.776	0.09175	158.0	23117	23275	20398

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Table 26: Shear wall forces with the rigid diaphragm assumption - Level 1

Wall	T_x [lbf-ft]	T_y [lbf-ft]	d_x [ft]	d_y [ft]	R [-]	F_t [lbf]	F_v [lbf]	F_{tot} [lbf]	F_{flex} [lbf]
A-2	1270000.0	3606000.0	86.19	0.0	0.07675	2388	21521	24822	24822
A-2.4	1270000.0	3606000.0	78.02	0.0	0.09162	2580	25690	28491	28491
A-3	1270000.0	3606000.0	68.27	0.0	0.04849	1152	13107	14259	10754
A-3.5	1270000.0	3606000.0	54.35	0.0	0.06265	1239	17713	18952	17611
A-3.7	1270000.0	3606000.0	50.77	0.0	0.03136	620.0	9487	10107	7996
A-4	1270000.0	3606000.0	41.94	0.0	0.03589	518.3	9601	10119	9160
A-5	1270000.0	3606000.0	20.60	0.0	0.05938	395.9	14927	15323	13967
A-5.1	1270000.0	3606000.0	20.02	0.0	0.06374	476.1	18475	18951	17914
A-6	1201000.0	3147000.0	4.064	0.0	0.06415	92.56	18715	18807	17914
A-6.1	1201000.0	3147000.0	4.648	0.0	0.06104	90.72	16041	16131	14979
A-6.8	1201000.0	3147000.0	29.31	0.0	0.06216	606.7	17010	17616	16422
A-7	1201000.0	3147000.0	38.40	0.0	0.06383	868.3	18585	19453	18345
A-7.8	1201000.0	3147000.0	58.73	0.0	0.03913	820.0	11474	12294	11614
A-8	1201000.0	3147000.0	64.48	0.0	0.05920	1161	14794	15955	14397
A-8.2	1201000.0	3147000.0	67.56	0.0	0.03144	781.1	9501	10282	8122
A-8.9	1201000.0	3147000.0	85.48	0.0	0.07663	2235	21487	24518	24518
A-9	1201000.0	3147000.0	85.81	0.0	0.07255	2215	21216	23431	22317
A-B-A-1	1735000.0	3282000.0	0.0	2.920	0.09051	208.5	26719	26928	21766
A-B-A-2.5	1735000.0	3282000.0	0.0	2.920	0.04299	91.84	11771	14148	14148
A-B-A-3.5	1735000.0	3282000.0	0.0	2.920	0.02622	56.01	7180	9795	9795
A-B-A-4	1735000.0	3282000.0	0.0	2.920	0.03870	82.68	10597	13060	13060
A-B-A-4.8	1735000.0	3282000.0	0.0	2.920	0.03309	70.69	9060	11612	11612
A-B-A-5.5	1735000.0	3282000.0	0.0	2.920	0.1398	313.2	40145	40458	31561
A-B-A-6.5	1735000.0	3282000.0	0.0	2.920	0.04299	91.84	11771	14148	14148
A-B-A-7.9	1735000.0	3282000.0	0.0	2.920	0.03730	79.68	10213	12700	12700
A-B-A-8.8	1735000.0	3282000.0	0.0	2.920	0.03031	64.74	8298	10883	10883
A-C-A-1.8	1640000.0	3282000.0	0.0	2.580	0.05477	98.88	14342	15483	15483
A-C-A-3.2	1640000.0	3282000.0	0.0	2.580	0.02666	50.34	7301	9038	9038
A-C-A-3.8	1640000.0	3282000.0	0.0	2.580	0.05807	103.7	15045	16208	16208
A-C-A-4.6	1640000.0	3282000.0	0.0	2.580	0.05477	98.88	14342	15483	15483
A-C-A-5.5	1640000.0	3282000.0	0.0	2.580	0.05041	92.38	13399	14515	14515
A-C-A-6.5	1640000.0	3282000.0	0.0	2.580	0.1097	221.9	32180	32402	25159
A-C-A-6.9	1640000.0	3282000.0	0.0	2.580	0.03549	67.00	9718	11128	11128
A-C-A-8.5	1640000.0	3282000.0	0.0	2.580	0.02831	53.45	7752	9435	9435
A-C-A-9.2	1640000.0	3282000.0	0.0	2.580	0.09994	203.5	29510	29713	23224

3.2.3 Final Shear Wall Nailing Selection

With the envelope procedure complete, we can now select the shear wall sheathing and nailing specifications.

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Table 27: Final shear wall design values - Level 4

Wall	$F_{x,cum}$ [lbf]	l_{wall} [ft]	v [plf]	$0.7v$ [plf]	v_{cap} [plf]	sides	s_{nail} [in]
A-2	7599	29.8	255	179	311	1.00	6.00
A-2.4	8722	29.8	293	205	311	1.00	6.00
A-3	4223	29.8	142	99.4	311	1.00	6.00
A-3.5	6023	29.8	202	142	311	1.00	6.00
A-3.7	2845	20.0	142	99.6	311	1.00	6.00
A-4	3123	20.0	156	109	311	1.00	6.00
A-5	4947	29.8	166	116	311	1.00	6.00
A-5.1	5971	29.8	201	141	311	1.00	6.00
A-6	5901	29.8	198	139	311	1.00	6.00
A-6.1	5149	29.8	173	121	311	1.00	6.00
A-6.8	5594	29.8	188	132	311	1.00	6.00
A-7	6119	29.8	206	144	311	1.00	6.00
A-7.8	3661	20.0	183	128	311	1.00	6.00
A-8	5158	29.8	173	121	311	1.00	6.00
A-8.2	2888	20.0	144	101	311	1.00	6.00
A-8.9	7506	29.8	252	177	311	1.00	6.00
A-9	7135	29.8	240	168	311	1.00	6.00
A-B-A-1	7111	20.0	356	249	311	1.00	6.00
A-B-A-2.5	4331	13.0	333	233	311	1.00	6.00
A-B-A-3.5	2998	9.00	333	233	311	1.00	6.00
A-B-A-4	3998	12.0	333	233	311	1.00	6.00
A-B-A-4.8	3555	10.7	333	233	311	1.00	6.00
A-B-A-5.5	10498	29.0	362	253	311	1.00	6.00
A-B-A-6.5	4331	13.0	333	233	311	1.00	6.00
A-B-A-7.9	3888	11.7	333	233	311	1.00	6.00
A-B-A-8.8	3332	10.0	333	233	311	1.00	6.00
A-C-A-1.8	5076	16.0	317	222	311	1.00	6.00
A-C-A-3.2	2767	9.34	296	207	311	1.00	6.00
A-C-A-3.8	5343	16.8	319	223	311	1.00	6.00
A-C-A-4.6	5076	16.0	317	222	311	1.00	6.00
A-C-A-5.5	4717	15.0	314	220	311	1.00	6.00
A-C-A-6.5	8534	26.0	328	230	311	1.00	6.00
A-C-A-6.9	3407	11.5	296	207	311	1.00	6.00
A-C-A-8.5	2888	9.75	296	207	311	1.00	6.00
A-C-A-9.2	7857	24.0	327	229	311	1.00	6.00

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Table 28: Final shear wall design values - Level 3

Wall	$F_{x,cum}$ [lbf]	l_{wall} [ft]	v [plf]	$0.7v$ [plf]	v_{cap} [plf]	sides	s_{nail} [in]
A-2	16570	29.8	557	390	461	1.00	4.00
A-2.4	18432	29.8	620	434	461	1.00	4.00
A-3	8377	29.8	282	197	311	1.00	6.00
A-3.5	12411	29.8	417	292	311	1.00	6.00
A-3.7	5980	20.0	299	209	311	1.00	6.00
A-4	6605	20.0	330	231	311	1.00	6.00
A-5	10076	29.8	339	237	311	1.00	6.00
A-5.1	12356	29.8	415	291	311	1.00	6.00
A-6	12086	29.8	406	284	311	1.00	6.00
A-6.1	10571	29.8	355	249	311	1.00	6.00
A-6.8	11518	29.8	387	271	311	1.00	6.00
A-7	12622	29.8	424	297	311	1.00	6.00
A-7.8	7761	20.0	388	272	311	1.00	6.00
A-8	10465	29.8	352	246	311	1.00	6.00
A-8.2	6057	20.0	303	212	311	1.00	6.00
A-8.9	16339	29.8	549	384	461	1.00	4.00
A-9	15673	29.8	527	369	461	1.00	4.00
A-B-A-1	14632	20.0	732	512	600	1.00	3.00
A-B-A-2.5	9153	13.0	704	493	600	1.00	3.00
A-B-A-3.5	6337	9.00	704	493	600	1.00	3.00
A-B-A-4	8449	12.0	704	493	600	1.00	3.00
A-B-A-4.8	7512	10.7	704	493	600	1.00	3.00
A-B-A-5.5	21377	29.0	737	516	600	1.00	3.00
A-B-A-6.5	9153	13.0	704	493	600	1.00	3.00
A-B-A-7.9	8216	11.7	704	493	600	1.00	3.00
A-B-A-8.8	7041	10.0	704	493	600	1.00	3.00
A-C-A-1.8	10313	16.0	645	451	461	1.00	4.00
A-C-A-3.2	5847	9.34	626	438	461	1.00	4.00
A-C-A-3.8	10827	16.8	646	452	461	1.00	4.00
A-C-A-4.6	10313	16.0	645	451	461	1.00	4.00
A-C-A-5.5	9608	15.0	641	448	461	1.00	4.00
A-C-A-6.5	18659	26.0	718	502	600	1.00	3.00
A-C-A-6.9	7199	11.5	626	438	461	1.00	4.00
A-C-A-8.5	6104	9.75	626	438	461	1.00	4.00
A-C-A-9.2	17174	24.0	716	501	600	1.00	3.00

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Table 29: Final shear wall design values - Level 2

Wall	$F_{x,cum}$ [lbf]	l_{wall} [ft]	v [plf]	$0.7v$ [plf]	v_{cap} [plf]	sides	s_{nail} [in]
A-2	21805	29.8	733	513	600	1.00	3.00
A-2.4	25025	29.8	841	589	600	1.00	3.00
A-3	12491	29.8	420	294	311	1.00	6.00
A-3.5	16541	29.8	556	389	461	1.00	4.00
A-3.7	7691	20.0	385	269	311	1.00	6.00
A-4	8531	20.0	427	299	311	1.00	6.00
A-5	15154	29.8	509	357	461	1.00	4.00
A-5.1	16492	29.8	554	388	461	1.00	4.00
A-6	16327	29.8	549	384	461	1.00	4.00
A-6.1	15921	29.8	535	375	461	1.00	4.00
A-6.8	15308	29.8	515	360	461	1.00	4.00
A-7	16876	29.8	567	397	461	1.00	4.00
A-7.8	10709	20.0	535	375	461	1.00	4.00
A-8	15750	29.8	529	371	461	1.00	4.00
A-8.2	7774	20.0	389	272	311	1.00	6.00
A-8.9	21535	29.8	724	507	600	1.00	3.00
A-9	20614	29.8	693	485	600	1.00	3.00
A-B-A-1	19118	20.0	956	669	770	1.00	2.00
A-B-A-2.5	12427	13.0	956	669	770	1.00	2.00
A-B-A-3.5	8603	9.00	956	669	770	1.00	2.00
A-B-A-4	11471	12.0	956	669	770	1.00	2.00
A-B-A-4.8	10199	10.7	956	669	770	1.00	2.00
A-B-A-5.5	37689	29.0	1300	910	921	2.00	4.00
A-B-A-6.5	12427	13.0	956	669	770	1.00	2.00
A-B-A-7.9	11155	11.7	956	669	770	1.00	2.00
A-B-A-8.8	9559	10.0	956	669	770	1.00	2.00
A-C-A-1.8	13599	16.0	850	595	600	1.00	3.00
A-C-A-3.2	7938	9.34	850	595	600	1.00	3.00
A-C-A-3.8	14236	16.8	850	595	600	1.00	3.00
A-C-A-4.6	13599	16.0	850	595	600	1.00	3.00
A-C-A-5.5	12749	15.0	850	595	600	1.00	3.00
A-C-A-6.5	25311	26.0	973	681	770	1.00	2.00
A-C-A-6.9	9774	11.5	850	595	600	1.00	3.00
A-C-A-8.5	8287	9.75	850	595	600	1.00	3.00
A-C-A-9.2	23275	24.0	970	679	770	1.00	2.00

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Table 30: Final shear wall design values - Level 1

Wall	$F_{x,cum}$ [lbf]	l_{wall} [ft]	v [plf]	$0.7v$ [plf]	v_{cap} [plf]	sides	s_{nail} [in]
A-2	24822	29.8	834	584	600	1.00	3.00
A-2.4	28491	29.8	958	670	770	1.00	2.00
A-3	14259	29.8	479	336	461	1.00	4.00
A-3.5	18952	29.8	637	446	461	1.00	4.00
A-3.7	10107	20.0	505	354	461	1.00	4.00
A-4	10119	20.0	506	354	461	1.00	4.00
A-5	15323	29.8	515	361	461	1.00	4.00
A-5.1	18951	29.8	637	446	461	1.00	4.00
A-6	18807	29.8	632	443	461	1.00	4.00
A-6.1	16131	29.8	542	380	461	1.00	4.00
A-6.8	17616	29.8	592	415	461	1.00	4.00
A-7	19453	29.8	654	458	461	1.00	4.00
A-7.8	12294	20.0	615	430	461	1.00	4.00
A-8	15955	29.8	536	375	461	1.00	4.00
A-8.2	10282	20.0	514	360	461	1.00	4.00
A-8.9	24518	29.8	824	577	600	1.00	3.00
A-9	23431	29.8	788	551	600	1.00	3.00
A-B-A-1	26928	20.0	1346	942	1200	2.00	3.00
A-B-A-2.5	14148	13.0	1088	762	770	1.00	2.00
A-B-A-3.5	9795	9.00	1088	762	770	1.00	2.00
A-B-A-4	13060	12.0	1088	762	770	1.00	2.00
A-B-A-4.8	11612	10.7	1088	762	770	1.00	2.00
A-B-A-5.5	40458	29.0	1395	977	1200	2.00	3.00
A-B-A-6.5	14148	13.0	1088	762	770	1.00	2.00
A-B-A-7.9	12700	11.7	1088	762	770	1.00	2.00
A-B-A-8.8	10883	10.0	1088	762	770	1.00	2.00
A-C-A-1.8	15483	16.0	968	677	770	1.00	2.00
A-C-A-3.2	9038	9.34	968	677	770	1.00	2.00
A-C-A-3.8	16208	16.8	968	677	770	1.00	2.00
A-C-A-4.6	15483	16.0	968	677	770	1.00	2.00
A-C-A-5.5	14515	15.0	968	677	770	1.00	2.00
A-C-A-6.5	32402	26.0	1246	872	921	2.00	4.00
A-C-A-6.9	11128	11.5	968	677	770	1.00	2.00
A-C-A-8.5	9435	9.75	968	677	770	1.00	2.00
A-C-A-9.2	29713	24.0	1238	867	921	2.00	4.00

3.2.4 Shear Wall Axial Forces

Tension Forces The shear forces get converted to overturning forces in the end posts of the shear walls. This overturning moment is counteracted by the dead load tributary to each shear wall according to ASCE 7-22 Sec. 2.4.5. The governing load combination equation is:

$$0.6D - 0.7E_v + 0.7E_h \quad (\text{ASCE 7-22 Eq. 2.4.5-10})$$

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Using the area dead loads from earlier, along with the tributary areas for each shear wall, we can get the dead load that is applied to each wall. This resists the overturning moment at the centroid of the wall.

For high-aspect-ratio walls, we can assume that the wall acts rigidly, thus:

$$M_R = D \frac{l_{wall}}{2} \quad (\text{units in kips and feet})$$

$$D = 0.6D_{nominal}$$

However, for longer (low aspect ratio) walls, the wall will not behave rigidly. We can only assume a portion of the vertical loading is transferred to the end posts; the rest is resolved through the wall studs. We will assume that a high aspect ratio is anything over 1:1; therefore, we will take the length of dead load being resisted as equal to the height of the wall.

$$M_R = \left(D * \min \left[\frac{h_{wall}}{l_{wall}}, 1 \right] \right) \frac{l_{wall}}{2} \quad (\text{units in kips and feet})$$

$$D = 0.6D_{nominal}$$

The vertical seismic effect is taken as, $E_v = 0.2S_{DS}D$ (units in kips), so the dead load length for the low aspect-ratio walls is also used.

$$P_v = 0.7(0.2S_{DS}D_{nominal})$$

The overturning moment is:

$$M_{OT} = Vh_{wall} + P_v \frac{l_{wall}}{2} \quad (\text{units in kips and feet})$$

$$V = 0.7V_{nominal}$$

We will assume that the holddowns are 6" inset from the end posts. The ASD tension caused by horizontal load effects counteracted by dead loads is:

$$T_{seismic} = \frac{M_{OT} - M_R}{l_{wall} - 0.5} \quad (\text{units in kips and feet})$$

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Table 31: End post tension forces - Level 4

Wall	l_{grav} [ft]	M_{Eh} [lbf-ft]	M_{Ev} [lbf-ft]	M_D [lbf-ft]	$T_{seismic}$ [lbf]
A-2	10.0	53191	4511	24164	1147
A-2.4	10.0	61053	2589	13868	1702
A-3	10.0	29559	3908	20938	428
A-3.5	10.0	42159	1600	8572	1203
A-3.7	10.0	19918	4323	23157	55.6
A-4	10.0	21863	4952	26528	14.7
A-5	10.0	34627	5076	27194	428
A-5.1	10.0	41799	3255	17440	944
A-6	10.0	41307	3255	17440	927
A-6.1	10.0	36043	5444	29165	421
A-6.8	10.0	39156	1492	7993	1116
A-7	10.0	42835	1667	8929	1216
A-7.8	10.0	25627	1570	8409	963
A-8	10.0	36109	2616	14016	845
A-8.2	10.0	20219	4391	23523	55.7
A-8.9	10.0	52540	4456	23869	1133
A-9	10.0	49945	2028	10863	1405
A-B-A-1	10.0	49779	0.0	0.0	2553
A-B-A-2.5	10.0	30317	0.0	0.0	2425
A-B-A-3.5	9.00	20989	0.0	0.0	2469
A-B-A-4	10.0	27985	0.0	0.0	2433
A-B-A-4.8	10.0	24883	0.0	0.0	2447
A-B-A-5.5	10.0	73485	0.0	0.0	2578
A-B-A-6.5	10.0	30317	0.0	0.0	2425
A-B-A-7.9	10.0	27215	0.0	0.0	2436
A-B-A-8.8	10.0	23321	0.0	0.0	2455
A-C-A-1.8	10.0	35530	0.0	0.0	2292
A-C-A-3.2	9.34	19367	0.0	0.0	2191
A-C-A-3.8	10.0	37401	0.0	0.0	2302
A-C-A-4.6	10.0	35530	0.0	0.0	2292
A-C-A-5.5	10.0	33019	0.0	0.0	2277
A-C-A-6.5	10.0	59737	0.0	0.0	2343
A-C-A-6.9	10.0	23846	0.0	0.0	2168
A-C-A-8.5	9.75	20217	0.0	0.0	2186
A-C-A-9.2	10.0	55002	0.0	0.0	2341

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Table 32: End post tension forces - Level 3

Wall	l_{grav} [ft]	M_{Eh} [lbf-ft]	M_{Ev} [lbf-ft]	M_D [lbf-ft]	$T_{seismic}$ [lbf]
A-2	10.5	121787	4973	26641	4569
A-2.4	10.5	135476	2854	15289	5908
A-3	10.5	61572	4309	23084	1892
A-3.5	10.5	91223	1764	9451	4059
A-3.7	10.5	43956	4766	25530	1245
A-4	10.5	48547	5459	29247	1284
A-5	10.5	74057	5597	29982	2126
A-5.1	10.5	90815	3589	19227	3514
A-6	10.5	88834	3589	19227	3430
A-6.1	10.5	77700	6002	32154	2184
A-6.8	10.5	84654	1645	8813	3766
A-7	10.5	92770	1838	9844	4114
A-7.8	10.5	57043	1731	9271	3502
A-8	10.5	76916	2884	15452	3045
A-8.2	10.5	44522	4841	25934	1257
A-8.9	10.5	120090	4912	26315	4506
A-9	10.5	115194	2236	11976	5011
A-B-A-1	10.5	107543	0.0	0.0	8068
A-B-A-2.5	10.5	67274	0.0	0.0	7807
A-B-A-3.5	9.00	46574	0.0	0.0	7949
A-B-A-4	10.5	62099	0.0	0.0	7833
A-B-A-4.8	10.5	55216	0.0	0.0	7876
A-B-A-5.5	10.5	157118	0.0	0.0	8091
A-B-A-6.5	10.5	67274	0.0	0.0	7807
A-B-A-7.9	10.5	60391	0.0	0.0	7843
A-B-A-8.8	10.0	51749	0.0	0.0	7902
A-C-A-1.8	10.5	75798	0.0	0.0	7182
A-C-A-3.2	9.34	42976	0.0	0.0	7052
A-C-A-3.8	10.5	79575	0.0	0.0	7199
A-C-A-4.6	10.5	75798	0.0	0.0	7182
A-C-A-5.5	10.5	70615	0.0	0.0	7147
A-C-A-6.5	10.5	137141	0.0	0.0	7721
A-C-A-6.9	10.5	52914	0.0	0.0	6978
A-C-A-8.5	9.75	44862	0.0	0.0	7036
A-C-A-9.2	10.5	126232	0.0	0.0	7712

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Table 33: End post tension forces - Level 2

Wall	l_{grav} [ft]	M_{Eh} [lbf-ft]	M_{Ev} [lbf-ft]	M_D [lbf-ft]	$T_{seismic}$ [lbf]
A-2	10.5	160264	4973	26641	9308
A-2.4	10.5	183931	2854	15289	11771
A-3	10.5	91809	4309	23084	4388
A-3.5	10.5	121579	1764	9451	7953
A-3.7	10.5	56530	4766	25530	3079
A-4	10.5	62701	5459	29247	3280
A-5	10.5	111381	5597	29982	5100
A-5.1	10.5	121215	3589	19227	7124
A-6	10.5	120003	3589	19227	6998
A-6.1	10.5	117018	6002	32154	5290
A-6.8	10.5	112510	1645	8813	7367
A-7	10.5	124039	1838	9844	8081
A-7.8	10.5	78714	1731	9271	7152
A-8	10.5	115762	2884	15452	6573
A-8.2	10.5	57138	4841	25934	3106
A-8.9	10.5	158285	4912	26315	9186
A-9	10.5	151515	2236	11976	9858
A-B-A-1	10.5	140515	0.0	0.0	15274
A-B-A-2.5	10.5	91335	0.0	0.0	15114
A-B-A-3.5	9.00	63232	0.0	0.0	15388
A-B-A-4	10.5	84309	0.0	0.0	15165
A-B-A-4.8	10.5	74965	0.0	0.0	15247
A-B-A-5.5	10.5	277013	0.0	0.0	17811
A-B-A-6.5	10.5	91335	0.0	0.0	15114
A-B-A-7.9	10.5	81991	0.0	0.0	15183
A-B-A-8.8	10.0	70258	0.0	0.0	15298
A-C-A-1.8	10.5	99951	0.0	0.0	13631
A-C-A-3.2	9.34	58347	0.0	0.0	13653
A-C-A-3.8	10.5	104637	0.0	0.0	13638
A-C-A-4.6	10.5	99951	0.0	0.0	13631
A-C-A-5.5	10.5	93704	0.0	0.0	13610
A-C-A-6.5	10.5	186034	0.0	0.0	15016
A-C-A-6.9	10.5	71840	0.0	0.0	13509
A-C-A-8.5	9.75	60908	0.0	0.0	13620
A-C-A-9.2	10.5	171073	0.0	0.0	14992

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Table 34: End post tension forces - Level 1

Wall	l_{grav} [ft]	M_{Eh} [lbf-ft]	M_{Ev} [lbf-ft]	M_D [lbf-ft]	$T_{seismic}$ [lbf]
A-2	11.5	199818	5965	31957	15250
A-2.4	11.5	229352	3424	18340	19102
A-3	11.5	114785	5169	27690	7543
A-3.5	11.5	152567	2116	11337	12853
A-3.7	11.5	81364	5717	30625	5974
A-4	11.5	81462	6549	35083	5994
A-5	11.5	123353	6713	35964	8317
A-5.1	11.5	152553	4305	23064	11698
A-6	11.5	151400	4305	23064	11532
A-6.1	11.5	129857	7200	38570	8657
A-6.8	11.5	141811	1973	10571	11921
A-7	11.5	156598	2204	11809	13106
A-7.8	11.5	98968	2076	11121	11763
A-8	11.5	128437	3460	18536	10448
A-8.2	11.5	82768	5807	31110	6053
A-8.9	11.5	197373	5892	31566	15056
A-9	11.5	188620	2682	14366	15907
A-B-A-1	11.5	216768	0.0	0.0	26390
A-B-A-2.5	11.5	113890	0.0	0.0	24225
A-B-A-3.5	9.00	78847	0.0	0.0	24664
A-B-A-4	11.5	105129	0.0	0.0	24306
A-B-A-4.8	10.7	93477	0.0	0.0	24439
A-B-A-5.5	11.5	325689	0.0	0.0	29239
A-B-A-6.5	11.5	113890	0.0	0.0	24225
A-B-A-7.9	11.5	102238	0.0	0.0	24336
A-B-A-8.8	10.0	87608	0.0	0.0	24519
A-C-A-1.8	11.5	124634	0.0	0.0	21672
A-C-A-3.2	9.34	72755	0.0	0.0	21883
A-C-A-3.8	11.5	130476	0.0	0.0	21667
A-C-A-4.6	11.5	124634	0.0	0.0	21672
A-C-A-5.5	11.5	116845	0.0	0.0	21668
A-C-A-6.5	11.5	260837	0.0	0.0	25245
A-C-A-6.9	11.5	89581	0.0	0.0	21653
A-C-A-8.5	9.75	75949	0.0	0.0	21831
A-C-A-9.2	11.5	239193	0.0	0.0	25170

Compression Forces We will calculate the ASD compression forces in a very similar way.

One significant change will be the additional gravity loads (D, L, and S). The bearing studs in the shear wall are designed to take the compressive gravity load across the length of the wall, so the tributary width for the gravity loads into the compression stud will come from the width of one stud bay (i.e., the end stud bay tributary to the end post in compression).

The governing load combination for compression is one of the following:



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$$1.0D + 0.7E_v + 0.7E_h \quad (\text{ASCE 7-22 Eq. 2.4.5-8})$$

$$1.0D + 0.525E_v + 0.525E_h + 0.75L + 0.1S \quad (\text{ASCE 7-22 Eq. 2.4.5-9})$$

The vertical seismic effect is taken as, $E_v = 0.2S_{DS}D$.

The live load is taken as $L = 40$ psf, while the snow load is taken as $S = 25$ psf.

Table 35: End post compression forces - Level 4

Wall	l_{grav} [ft]	M_{eq8} [lbf-ft]	M_{eq9} [lbf-ft]	$T_{seismic}$ [lbf]
A-2	1.33	53987	41334	1846
A-2.4	1.33	61510	46617	2103
A-3	1.33	30249	23418	1034
A-3.5	1.33	42441	32130	1451
A-3.7	1.33	20681	16319	1061
A-4	1.33	22737	17979	1166
A-5	1.33	35523	27592	1214
A-5.1	1.33	42374	32389	1449
A-6	1.33	41882	32020	1432
A-6.1	1.33	37004	28772	1265
A-6.8	1.33	39420	29844	1348
A-7	1.33	43129	32658	1474
A-7.8	1.33	25904	19722	1328
A-8	1.33	36571	27917	1250
A-8.2	1.33	20994	16567	1077
A-8.9	1.33	53327	40829	1823
A-9	1.33	50303	38107	1720
A-B-A-1	1.33	49779	37334	2553
A-B-A-2.5	1.33	30317	22738	2425
A-B-A-3.5	1.33	20989	15742	2469
A-B-A-4	1.33	27985	20989	2433
A-B-A-4.8	1.33	24883	18663	2447
A-B-A-5.5	1.33	73485	55113	2578
A-B-A-6.5	1.33	30317	22738	2425
A-B-A-7.9	1.33	27215	20412	2436
A-B-A-8.8	1.33	23321	17491	2455
A-C-A-1.8	1.33	35530	26648	2292
A-C-A-3.2	1.33	19367	14525	2191
A-C-A-3.8	1.33	37401	28051	2302
A-C-A-4.6	1.33	35530	26648	2292
A-C-A-5.5	1.33	33019	24764	2277
A-C-A-6.5	1.33	59737	44802	2343
A-C-A-6.9	1.33	23846	17885	2168
A-C-A-8.5	1.33	20217	15163	2186
A-C-A-9.2	1.33	55002	41252	2341

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Table 36: End post compression forces - Level 3

Wall	l_{grav} [ft]	M_{eq8} [lbf-ft]	M_{eq9} [lbf-ft]	$T_{seismic}$ [lbf]
A-2	1.33	122583	92781	6037
A-2.4	1.33	135933	102434	6750
A-3	1.33	62261	47427	3163
A-3.5	1.33	91505	68928	4579
A-3.7	1.33	44719	34348	3354
A-4	1.33	49421	37992	3700
A-5	1.33	74953	57164	3777
A-5.1	1.33	91389	69151	4573
A-6	1.33	89409	67665	4489
A-6.1	1.33	78661	60014	3954
A-6.8	1.33	84917	63967	4251
A-7	1.33	93064	70110	4656
A-7.8	1.33	57320	43283	4268
A-8	1.33	77378	58523	3896
A-8.2	1.33	45297	34794	3400
A-8.9	1.33	120877	91491	5956
A-9	1.33	115552	87043	5670
A-B-A-1	1.33	107543	80658	8068
A-B-A-2.5	1.33	67274	50455	7807
A-B-A-3.5	1.33	46574	34930	7949
A-B-A-4	1.33	62099	46574	7833
A-B-A-4.8	1.33	55216	41412	7876
A-B-A-5.5	1.33	157118	117838	8091
A-B-A-6.5	1.33	67274	50455	7807
A-B-A-7.9	1.33	60391	45293	7843
A-B-A-8.8	1.33	51749	38812	7902
A-C-A-1.8	1.33	75798	56849	7182
A-C-A-3.2	1.33	42976	32232	7052
A-C-A-3.8	1.33	79575	59682	7199
A-C-A-4.6	1.33	75798	56849	7182
A-C-A-5.5	1.33	70615	52961	7147
A-C-A-6.5	1.33	137141	102856	7721
A-C-A-6.9	1.33	52914	39686	6978
A-C-A-8.5	1.33	44862	33647	7036
A-C-A-9.2	1.33	126232	94674	7712

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Table 37: End post compression forces - Level 2

Wall	l_{grav} [ft]	M_{eq8} [lbf-ft]	M_{eq9} [lbf-ft]	$T_{seismic}$ [lbf]
A-2	1.33	161060	121639	11543
A-2.4	1.33	184388	138775	13054
A-3	1.33	92499	70105	6325
A-3.5	1.33	121861	91695	8746
A-3.7	1.33	57293	43778	6292
A-4	1.33	63575	48607	6961
A-5	1.33	112277	85157	7615
A-5.1	1.33	121789	91951	8737
A-6	1.33	120578	91042	8611
A-6.1	1.33	117979	89502	7988
A-6.8	1.33	112774	84859	8106
A-7	1.33	124333	93562	8907
A-7.8	1.33	78991	59537	8319
A-8	1.33	116224	87658	7869
A-8.2	1.33	57913	44256	6369
A-8.9	1.33	159072	120137	11394
A-9	1.33	151873	114284	10862
A-B-A-1	1.33	140515	105386	15274
A-B-A-2.5	1.33	91335	68501	15114
A-B-A-3.5	1.33	63232	47424	15388
A-B-A-4	1.33	84309	63232	15165
A-B-A-4.8	1.33	74965	56224	15247
A-B-A-5.5	1.33	277013	207759	17811
A-B-A-6.5	1.33	91335	68501	15114
A-B-A-7.9	1.33	81991	61493	15183
A-B-A-8.8	1.33	70258	52693	15298
A-C-A-1.8	1.33	99951	74964	13631
A-C-A-3.2	1.33	58347	43760	13653
A-C-A-3.8	1.33	104637	78477	13638
A-C-A-4.6	1.33	99951	74964	13631
A-C-A-5.5	1.33	93704	70278	13610
A-C-A-6.5	1.33	186034	139525	15016
A-C-A-6.9	1.33	71840	53880	13509
A-C-A-8.5	1.33	60908	45681	13620
A-C-A-9.2	1.33	171073	128305	14992

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Table 38: End post compression forces - Level 1

Wall	l_{grav} [ft]	M_{eq8} [lbf-ft]	M_{eq9} [lbf-ft]	$T_{seismic}$ [lbf]
A-2	1.33	200614	151304	18401
A-2.4	1.33	229809	172841	20911
A-3	1.33	115475	87337	10273
A-3.5	1.33	152850	114937	13971
A-3.7	1.33	82127	62404	10504
A-4	1.33	82336	62678	11183
A-5	1.33	124249	94137	11863
A-5.1	1.33	153127	115455	13972
A-6	1.33	151975	114590	13807
A-6.1	1.33	130818	99132	12460
A-6.8	1.33	142075	106835	12964
A-7	1.33	156892	117981	14271
A-7.8	1.33	99245	74728	13408
A-8	1.33	128899	97164	12276
A-8.2	1.33	83543	63479	10654
A-8.9	1.33	198160	149453	18169
A-9	1.33	188978	142113	17323
A-B-A-1	1.33	216768	162576	26390
A-B-A-2.5	1.33	113890	85417	24225
A-B-A-3.5	1.33	78847	59135	24664
A-B-A-4	1.33	105129	78847	24306
A-B-A-4.8	1.33	93477	70108	24439
A-B-A-5.5	1.33	325689	244267	29239
A-B-A-6.5	1.33	113890	85417	24225
A-B-A-7.9	1.33	102238	76679	24336
A-B-A-8.8	1.33	87608	65706	24519
A-C-A-1.8	1.33	124634	93476	21672
A-C-A-3.2	1.33	72755	54566	21883
A-C-A-3.8	1.33	130476	97857	21667
A-C-A-4.6	1.33	124634	93476	21672
A-C-A-5.5	1.33	116845	87633	21668
A-C-A-6.5	1.33	260837	195628	25245
A-C-A-6.9	1.33	89581	67186	21653
A-C-A-8.5	1.33	75949	56962	21831
A-C-A-9.2	1.33	239193	179395	25170

3.3 Diaphragm Design

We must first get the diaphragm inertial design forces. These are distinct from the seismic forces determined earlier.

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$$F_{px} = \frac{\sum_{i=x}^n F_i}{\sum_{i=x}^n w_i} w_{px} \quad (\text{ASCE Eq. 12.10-1})$$

$$F_{px} \geq 0.2 S_{DS} I_e w_{px} \quad (\text{ASCE Eq. 12.10-2})$$

$$F_{px} \leq 0.4 S_{DS} I_e w_{px} \quad (\text{ASCE Eq. 12.10-3})$$

These forces are shown in [Table 40](#)

Table 39: Diaphragm Forces - x

Level	w_x [kip]	$\sum F_x$ [kip]	$F_{px,u}$ [kip]	$F_{px,max}$ [kip]	F_{px} [kip]
5	421	85.5	74.9	384	74.9
4	614	181	109	560	109
3	615	245	109	561	109
2	619	279	110	565	110
1	0.0	279	0.0	0.0	0.0

Table 40: Diaphragm Forces - y

Level	w_x [kip]	$\sum F_x$ [kip]	$F_{px,u}$ [kip]	$F_{px,max}$ [kip]	F_{px} [kip]
5	421	85.5	74.9	110	74.9
4	614	181	109	160	109
3	615	245	109	160	109
2	619	279	110	161	110
1	0.0	279	0.0	0.0	0.0

With these diaphragm forces, we can design the diaphragm for the worst-case cantilever and simply-supported conditions.

The longest simply-supported span in the N-S direction is about 34.5 feet, while the longest cantilever span is about 32 feet. The depth of the diaphragm in the N-S direction is 67 feet. The longest simply-supported span in the E-W direction is about 5.5 feet, while the longest cantilever span is about 31 feet. The depth of the diaphragm in the E-W direction is 230 feet.

Diaphragm unit shear demands are shown in [Table 41](#)

We will assume the diaphragm sheathing has the following properties:

- Unblocked
- WSP Sheathing
- 15/32" thick
- 10d common nails
- Plywood
- Nailed face is 2"
- Edge adjoining cases 2 through 6

The diaphragm capacity checks for this chosen sheathing are shown in [Table 42](#)

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Table 41: Diaphragm Unit Shear Demands

Level	Direction	F_{px} [lbf]	w [psf]	V_{simp} [lbf]	V_{cant} [lbf]	v_{simp} [plf]	v_{cant} [plf]
5	N-S	74932	4.86	5620	10425	83.9	156
	E-W	74932	4.86	3076	34670	13.4	151
4	N-S	109092	7.08	8182	15178	122	227
	E-W	109092	7.08	4478	50475	19.5	219
3	N-S	109414	7.10	8206	15223	122	227
	E-W	109414	7.10	4491	50624	19.5	220
2	N-S	110058	7.14	8254	15312	123	229
	E-W	110058	7.14	4517	50923	19.6	221

Table 42: Diaphragm Capacity Checks

Level	Direction	v_{simp} [plf]	$v_{cap,simp}$ [plf]	DCR_{simp} [-]	v_{cant} [plf]	$v_{cap,cant}$ [plf]	DCR_{cant} [-]
5	N-S	83.9	189	0.443	156	189	0.822
	E-W	13.4	189	0.0706	151	189	0.796
4	N-S	122	189	0.645	227	255	0.887
	E-W	19.5	189	0.103	219	255	0.859
3	N-S	122	189	0.647	227	255	0.890
	E-W	19.5	189	0.103	220	255	0.862
2	N-S	123	189	0.651	229	255	0.895
	E-W	19.6	189	0.104	221	255	0.867

∴ All shear capacities are OK

3.4 Story Drift Determination

The inelastic design story drift is calculated as the difference in deflections between stories at their centers of mass.

We can get the elastic story drift at the center of mass by simply dividing the total story shear by the total wall stiffness in the relevant direction. Then, we can use the equation for inelastic story drift from ASCE 7.

$$\delta_{xe} = F/k$$

$$\delta_{DE} = \frac{C_d \delta_{xe}}{I_e} + \delta_{di}$$

For buildings with a torsional irregularity ratio of over 1.2, we must calculate the accidental torsional amplification factor, $A_{x\tau}$, in each direction, then amplify the accidental moment by that factor. All shear design forces values are recalculated with this value.

We must also look at the edges of the rigid diaphragm for these buildings with the $TIR > 2$. The story drift for these buildings is the greatest difference between the edges of the structure at each level and not just at the center of mass. There are 8 different possibilities that will be simplified into one number (max):

1. far left edge, counter-clockwise rotation, EW forces



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2. far right edge, counter-clockwise rotation, EW forces
3. far left edge, counter-clockwise rotation, NS forces
4. far right edge, counter-clockwise rotation, NS forces
5. far left edge, clockwise rotation, EW forces
6. far right edge, clockwise rotation, EW forces
7. far left edge, clockwise rotation, NS forces
8. far right edge, clockwise rotation, NS forces

Note: it is taken into account that $TIR > 1.4$ in both directions results in $p = 1.3$. The value of p is increased to 1.3 when necessary.

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Table 43: Story Drifts - Level 4

Wall	Direction	TIR [-]	A_x [-]	A_y [-]	$\max \delta_{elastic}$ [ft]	$\max \delta_{DE}$ [ft]	Δ_{allow} [ft]
A-2	y	1.42	1.07	1.39	0.0	0.0	0.200
A-2.4	y	1.42	1.07	1.39	0.0	0.0	0.200
A-3	y	1.42	1.07	1.39	0.0	0.0	0.200
A-3.5	y	1.42	1.07	1.39	0.0	0.0	0.200
A-3.7	y	1.42	1.07	1.39	0.0	0.0	0.200
A-4	y	1.42	1.07	1.39	0.0	0.0	0.200
A-5	y	1.42	1.07	1.39	0.0	0.0	0.200
A-5.1	y	1.42	1.07	1.39	0.0	0.0	0.200
A-6	y	1.34	1.00	1.25	0.0	0.0	0.200
A-6.1	y	1.34	1.00	1.25	0.0	0.0	0.200
A-6.8	y	1.34	1.00	1.25	0.0	0.0	0.200
A-7	y	1.34	1.00	1.25	0.0	0.0	0.200
A-7.8	y	1.34	1.00	1.25	0.0	0.0	0.200
A-8	y	1.34	1.00	1.25	0.0	0.0	0.200
A-8.2	y	1.34	1.00	1.25	0.0	0.0	0.200
A-8.9	y	1.34	1.00	1.25	0.0	0.0	0.200
A-9	y	1.34	1.00	1.25	0.0	0.0	0.200
A-B-A-1	x	1.45	1.00	1.46	0.0	0.0	0.200
A-B-A-2.5	x	1.45	1.00	1.46	0.0	0.0	0.200
A-B-A-3.5	x	1.45	1.00	1.46	0.0	0.0	0.200
A-B-A-4	x	1.45	1.00	1.46	0.0	0.0	0.200
A-B-A-4.8	x	1.45	1.00	1.46	0.0	0.0	0.200
A-B-A-5.5	x	1.45	1.00	1.46	0.0	0.0	0.200
A-B-A-6.5	x	1.45	1.00	1.46	0.0	0.0	0.200
A-B-A-7.9	x	1.45	1.00	1.46	0.0	0.0	0.200
A-B-A-8.8	x	1.45	1.00	1.46	0.0	0.0	0.200
A-C-A-1.8	x	1.45	1.00	1.46	0.0	0.0	0.200
A-C-A-3.2	x	1.45	1.00	1.46	0.0	0.0	0.200
A-C-A-3.8	x	1.45	1.00	1.46	0.0	0.0	0.200
A-C-A-4.6	x	1.45	1.00	1.46	0.0	0.0	0.200
A-C-A-5.5	x	1.45	1.00	1.46	0.0	0.0	0.200
A-C-A-6.5	x	1.45	1.00	1.46	0.0	0.0	0.200
A-C-A-6.9	x	1.45	1.00	1.46	0.0	0.0	0.200
A-C-A-8.5	x	1.45	1.00	1.46	0.0	0.0	0.200
A-C-A-9.2	x	1.45	1.00	1.46	0.0	0.0	0.200

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Table 44: Story Drifts - Level 3

Wall	Direction	TIR [-]	A_x [-]	A_y [-]	$\max \delta_{elastic}$ [ft]	$\max \delta_{DE}$ [ft]	Δ_{allow} [ft]
A-2	y	1.40	1.05	1.36	0.0164	0.0658	0.210
A-2.4	y	1.40	1.05	1.36	0.0164	0.0658	0.210
A-3	y	1.40	1.05	1.36	0.0164	0.0658	0.210
A-3.5	y	1.40	1.05	1.36	0.0164	0.0658	0.210
A-3.7	y	1.40	1.05	1.36	0.0164	0.0658	0.210
A-4	y	1.40	1.05	1.36	0.0164	0.0658	0.210
A-5	y	1.40	1.05	1.36	0.0164	0.0658	0.210
A-5.1	y	1.40	1.05	1.36	0.0164	0.0658	0.210
A-6	y	1.33	1.00	1.23	0.0135	0.0539	0.210
A-6.1	y	1.33	1.00	1.23	0.0135	0.0539	0.210
A-6.8	y	1.33	1.00	1.23	0.0135	0.0539	0.210
A-7	y	1.33	1.00	1.23	0.0135	0.0539	0.210
A-7.8	y	1.33	1.00	1.23	0.0135	0.0539	0.210
A-8	y	1.33	1.00	1.23	0.0135	0.0539	0.210
A-8.2	y	1.33	1.00	1.23	0.0135	0.0539	0.210
A-8.9	y	1.33	1.00	1.23	0.0135	0.0539	0.210
A-9	y	1.33	1.00	1.23	0.0135	0.0539	0.210
A-B-A-1	x	1.51	1.00	1.58	0.0150	0.0601	0.210
A-B-A-2.5	x	1.51	1.00	1.58	0.0150	0.0601	0.210
A-B-A-3.5	x	1.51	1.00	1.58	0.0150	0.0601	0.210
A-B-A-4	x	1.51	1.00	1.58	0.0150	0.0601	0.210
A-B-A-4.8	x	1.51	1.00	1.58	0.0150	0.0601	0.210
A-B-A-5.5	x	1.51	1.00	1.58	0.0150	0.0601	0.210
A-B-A-6.5	x	1.51	1.00	1.58	0.0150	0.0601	0.210
A-B-A-7.9	x	1.51	1.00	1.58	0.0150	0.0601	0.210
A-B-A-8.8	x	1.51	1.00	1.58	0.0150	0.0601	0.210
A-C-A-1.8	x	1.51	1.00	1.58	0.0150	0.0601	0.210
A-C-A-3.2	x	1.51	1.00	1.58	0.0150	0.0601	0.210
A-C-A-3.8	x	1.51	1.00	1.58	0.0150	0.0601	0.210
A-C-A-4.6	x	1.51	1.00	1.58	0.0150	0.0601	0.210
A-C-A-5.5	x	1.51	1.00	1.58	0.0150	0.0601	0.210
A-C-A-6.5	x	1.51	1.00	1.58	0.0150	0.0601	0.210
A-C-A-6.9	x	1.51	1.00	1.58	0.0150	0.0601	0.210
A-C-A-8.5	x	1.51	1.00	1.58	0.0150	0.0601	0.210
A-C-A-9.2	x	1.51	1.00	1.58	0.0150	0.0601	0.210

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Table 45: Story Drifts - Level 2

Wall	Direction	TIR [-]	A_x [-]	A_y [-]	$\max \delta_{elastic}$ [ft]	$\max \delta_{DE}$ [ft]	Δ_{allow} [ft]
A-2	y	1.43	1.08	1.41	0.00600	0.0240	0.210
A-2.4	y	1.43	1.08	1.41	0.00600	0.0240	0.210
A-3	y	1.43	1.08	1.41	0.00600	0.0240	0.210
A-3.5	y	1.43	1.08	1.41	0.00600	0.0240	0.210
A-3.7	y	1.43	1.08	1.41	0.00600	0.0240	0.210
A-4	y	1.43	1.08	1.41	0.00600	0.0240	0.210
A-5	y	1.43	1.08	1.41	0.00600	0.0240	0.210
A-5.1	y	1.43	1.08	1.41	0.00600	0.0240	0.210
A-6	y	1.33	1.00	1.24	0.00445	0.0178	0.210
A-6.1	y	1.33	1.00	1.24	0.00445	0.0178	0.210
A-6.8	y	1.33	1.00	1.24	0.00445	0.0178	0.210
A-7	y	1.33	1.00	1.24	0.00445	0.0178	0.210
A-7.8	y	1.33	1.00	1.24	0.00445	0.0178	0.210
A-8	y	1.33	1.00	1.24	0.00445	0.0178	0.210
A-8.2	y	1.33	1.00	1.24	0.00445	0.0178	0.210
A-8.9	y	1.33	1.00	1.24	0.00445	0.0178	0.210
A-9	y	1.33	1.00	1.24	0.00445	0.0178	0.210
A-B-A-1	x	1.56	1.00	1.70	0.00328	0.0131	0.210
A-B-A-2.5	x	1.56	1.00	1.70	0.00328	0.0131	0.210
A-B-A-3.5	x	1.56	1.00	1.70	0.00328	0.0131	0.210
A-B-A-4	x	1.56	1.00	1.70	0.00328	0.0131	0.210
A-B-A-4.8	x	1.56	1.00	1.70	0.00328	0.0131	0.210
A-B-A-5.5	x	1.56	1.00	1.70	0.00328	0.0131	0.210
A-B-A-6.5	x	1.56	1.00	1.70	0.00328	0.0131	0.210
A-B-A-7.9	x	1.56	1.00	1.70	0.00328	0.0131	0.210
A-B-A-8.8	x	1.56	1.00	1.70	0.00328	0.0131	0.210
A-C-A-1.8	x	1.56	1.00	1.70	0.00328	0.0131	0.210
A-C-A-3.2	x	1.56	1.00	1.70	0.00328	0.0131	0.210
A-C-A-3.8	x	1.56	1.00	1.70	0.00328	0.0131	0.210
A-C-A-4.6	x	1.56	1.00	1.70	0.00328	0.0131	0.210
A-C-A-5.5	x	1.56	1.00	1.70	0.00328	0.0131	0.210
A-C-A-6.5	x	1.56	1.00	1.70	0.00328	0.0131	0.210
A-C-A-6.9	x	1.56	1.00	1.70	0.00328	0.0131	0.210
A-C-A-8.5	x	1.56	1.00	1.70	0.00328	0.0131	0.210
A-C-A-9.2	x	1.56	1.00	1.70	0.00328	0.0131	0.210

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Table 46: Story Drifts - Level 1

Wall	Direction	TIR [-]	A_x [-]	A_y [-]	$\max \delta_{elastic}$ [ft]	$\max \delta_{DE}$ [ft]	Δ_{allow} [ft]
A-2	y	1.40	1.05	1.36	0.00805	0.0322	0.230
A-2.4	y	1.40	1.05	1.36	0.00805	0.0322	0.230
A-3	y	1.40	1.05	1.36	0.00805	0.0322	0.230
A-3.5	y	1.40	1.05	1.36	0.00805	0.0322	0.230
A-3.7	y	1.40	1.05	1.36	0.00805	0.0322	0.230
A-4	y	1.40	1.05	1.36	0.00805	0.0322	0.230
A-5	y	1.40	1.05	1.36	0.00805	0.0322	0.230
A-5.1	y	1.40	1.05	1.36	0.00805	0.0322	0.230
A-6	y	1.36	1.01	1.28	0.00900	0.0360	0.230
A-6.1	y	1.36	1.01	1.28	0.00900	0.0360	0.230
A-6.8	y	1.36	1.01	1.28	0.00900	0.0360	0.230
A-7	y	1.36	1.01	1.28	0.00900	0.0360	0.230
A-7.8	y	1.36	1.01	1.28	0.00900	0.0360	0.230
A-8	y	1.36	1.01	1.28	0.00900	0.0360	0.230
A-8.2	y	1.36	1.01	1.28	0.00900	0.0360	0.230
A-8.9	y	1.36	1.01	1.28	0.00900	0.0360	0.230
A-9	y	1.36	1.01	1.28	0.00900	0.0360	0.230
A-B-A-1	x	1.61	1.00	1.80	0.00434	0.0174	0.230
A-B-A-2.5	x	1.61	1.00	1.80	0.00434	0.0174	0.230
A-B-A-3.5	x	1.61	1.00	1.80	0.00434	0.0174	0.230
A-B-A-4	x	1.61	1.00	1.80	0.00434	0.0174	0.230
A-B-A-4.8	x	1.61	1.00	1.80	0.00434	0.0174	0.230
A-B-A-5.5	x	1.61	1.00	1.80	0.00434	0.0174	0.230
A-B-A-6.5	x	1.61	1.00	1.80	0.00434	0.0174	0.230
A-B-A-7.9	x	1.61	1.00	1.80	0.00434	0.0174	0.230
A-B-A-8.8	x	1.61	1.00	1.80	0.00434	0.0174	0.230
A-C-A-1.8	x	1.61	1.00	1.80	0.00434	0.0174	0.230
A-C-A-3.2	x	1.61	1.00	1.80	0.00434	0.0174	0.230
A-C-A-3.8	x	1.61	1.00	1.80	0.00434	0.0174	0.230
A-C-A-4.6	x	1.61	1.00	1.80	0.00434	0.0174	0.230
A-C-A-5.5	x	1.61	1.00	1.80	0.00434	0.0174	0.230
A-C-A-6.5	x	1.61	1.00	1.80	0.00434	0.0174	0.230
A-C-A-6.9	x	1.61	1.00	1.80	0.00434	0.0174	0.230
A-C-A-8.5	x	1.61	1.00	1.80	0.00434	0.0174	0.230
A-C-A-9.2	x	1.61	1.00	1.80	0.00434	0.0174	0.230